

A Measure-Theoretic Discrete Curvature Framework for Metabolic Gene Sensitivity: From Active-Set Geometry to Transcriptional Response

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Abstract

When a cell’s environment changes, its metabolic network reroutes: the optimal fluxes move, and the map from environment to optimum is piecewise linear, not smooth. We measure how this map bends. Its second derivative, taken in the sense of distributions, is a matrix-valued Radon measure concentrated on the boundaries where the network switches between active constraint sets — the discrete curvature carrier of the network’s rerouting geometry. Parameter loops accumulate drift linearly in loop size (measured slope 1.00; a smooth map would scale quadratically), and 93.4–100.0% of second-order response mass lands on active-set switches in every boundary-crossing sweep. Under mesh refinement the atomic measure converges weakly to the smooth curvature density and the dual-cell reconstruction converges strongly in L^1 , while total-variation convergence fails in general; a resolution parameter σ separates the discrete and smooth regimes through the measured window $h \ll \sigma \ll L_{\text{var}}$ (L_{var} : smooth variation scale). From the measure we derive a gene-level sensitivity metric κ^μ . On *E. coli* (M3D microarray compendium, $n = 424$ evaluated genes), it predicts carbon-depletion transcriptional response ($r = +0.395$, $p = 2.6 \times 10^{-17}$; partial $r = +0.269$ with reference-level control); the association is robust to the tie-breaking convention and to random-panel sampling. The association does not propagate to the protein layer: cells transcribe rerouting potential while buffering its translation. Double-knockout epistasis aligns with active-set structure ($\rho_S = 0.865$), and genotype loops close with phenotypic memory (66% non-reverting). The framework unifies the atomic measure, its smooth coarse-grainings, and its trajectory integrals as one object at different resolutions.

Keywords: flux balance analysis; discrete curvature; active set; parametric linear programming; flux rerouting; epistasis

1 Introduction

The object. Let a metabolic network with m reactions be operated by flux balance analysis (Varma and Palsson, 1994; Orth et al., 2010) with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \rightarrow \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming (Borrelli et al., 2003)): as the parameters move, the optimum is a fixed linear function of them within each region of parameter space, and at the region boundaries the set of constraints that hold with equality (the *active set*) switches and the slope of v changes. These boundaries — the active-set strata of the constraint complex — are where the map bends. Its distributional second derivative $\mu = D^2v$ is therefore not a function but a *measure*, concentrated on the switching boundaries: the discrete curvature carrier of the network’s rerouting geometry.

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The claim structure. This manuscript establishes, in order:

- (i) the atom structure and weak convergence of μ_h under refinement (Theorem 3.1, §3), with an exact counterexample calculus delimiting what is provable (weak convergence, strong cell-wise reconstruction) from what fails (total-variation convergence);
- (ii) the two-layer separation: the codimension-one Hessian layer is the unbiased curvature estimator; the codimension-two angle-defect layer converges to the Gauss-map area density with an exact sec-law bias (Proposition 3.3);
- (iii) the regime dichotomy: piecewise-affine maps with fixed kink complexes produce $O(\varepsilon)$ loop holonomy (measured slope 1.00) where smooth maps produce $O(\varepsilon^2)$;
- (iv) the value function $\Phi = c_{\text{bio}}^\top v^*$ (the maximum achievable growth rate) is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes;
- (v) the measure-derived metric κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$), generalizes to oxygen-limitation and carbon-switch trajectories with matched responses ($r = +0.32, +0.22$; §5.5), is robust to the tie-break convention: $r \in [+0.386, +0.396]$ across five selection rules (§5.6), is buffered at the protein layer (§5), and its event measures stabilize at the Glivenko–Cantelli rate across random panels (§5.7).

Positioning. Kochanowski et al. (2013) demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

Plan of the paper. Section 2 defines the measure and the metric, develops the value-layer structure and the value–flux coupling, and records the categorical reading in brief. Section 3 states the refinement–resolution bridge. Section 4 reports the computational validation of the geometric claims, and Section 5 the empirical association and the protein-layer decision. Section 6 discusses limitations and scope. The Methods, a disambiguation table for colliding gene counts, and the proofs complete the paper.

2 The discrete curvature measure of parametric FBA

2.1 The atomic curvature measure

When the parameter path crosses a switching boundary, the optimal flux vector changes slope abruptly. The following definition collects all of these slope changes into a single object. Read u as the optimal-flux map and $\mathcal{C}$ as its active-set complex.

Definition 2.1 (Atomic curvature measure). Let $u : \Omega \rightarrow \mathbb{R}^m$ be continuous and piecewise affine on a polyhedral complex $\mathcal{C}$ covering Ω . For each codimension-one facet F of $\mathcal{C}$ shared by cells τ^- , τ^+ , let $[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-}$, let n_F be the unit conormal oriented $\tau^- \rightarrow \tau^+$, and let $|F|$ be the facet measure. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu = D^2 u = \sum_F A_F \delta_F, \quad A_F = [\nabla u]_F \otimes n_F |F| \in \mathbb{R}^{m \times p \times p}.$$

In words: each codimension-one facet F shared by two cells τ^- and τ^+ is one switching boundary; the jump matrix $[\nabla u]_F$ is the change of slope across it; n_F orients the jump, $|F|$ weights it by the size of the boundary, and δ_F places unit mass on the boundary itself. The atom A_F is thus the slope change, oriented and weighted by the boundary it lives on, and μ is the sum of all atoms — zero away from the switching boundaries.

Remark 2.2 (Support). μ is concentrated on the $(p-1)$ -skeleton. The $(p-2)$ -skeleton carries a different measure (the angle-defect layer, Proposition 3.3); the two must not be conflated. In $p=1$ the atoms are the slope jumps at breakpoints.

The measure-theoretic language is classical: matrix-valued Radon measures and their weak convergence (Billingsley, 1999), the Alexandrov Hessian of convex functions (Alexandrov, 1939), the Monge–Ampère measure (Gutiérrez, 2001), and Regge-type defect measures for piecewise-flat curvature (Regge, 1961; Cheeger et al., 1984) on the geometric side; chamber complexes, normal fans, and multi-parametric LP value functions (Rockafellar, 1970; Ziegler, 1995; Borrelli et al., 2003) on the polyhedral side. The Kantorovich–Rubinstein (flat) norm used in the refinement–resolution bridge is the standard optimal-transport flat norm (Villani, 2009).

2.2 The measure-theoretic sensitivity metric

The measure μ lives on parameter space; the biology enters when its mass is integrated along the trajectories that environments actually take, and the accumulated mass is attributed to genes.

Definition 2.3 (κ^μ). For a C^1 parameter trajectory $\theta(t)$, $t \in [0, T]$, and its piecewise-affine optimal-flux response $v(\theta(t))$, let $\{t_e\}_e$ be the event times at which the trajectory crosses active-set strata, with atoms $A_e = A_{F(e)}$ evaluated on the crossed facet. Define

$$\kappa^\mu(g) = \frac{1}{T} \sum_e \|A_e\| w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m \times p \times p}$ and $w_e(g) \in \{0, 1\}$ selects the events whose crossed stratum changes the activity of reactions assigned to gene g (max over its reactions, per the GPR association).

Remark 2.4 (κ^μ integrates along a one-dimensional path). κ^μ integrates measure mass along a one-dimensional trajectory — the regime in which total variation and mass are automatically consistent (see §3); its resolution independence (measured mass 288.77 stable under $4 \times / 8 \times$ trajectory refinement) is exactly the L^1 -reconstruction regime of Theorem 3.1(iv). Its per-reaction masses are the components of the flux-layer crease measure whose objective contraction is $D^2\Phi$ (Theorem 2.11): the metric is the primal layer of the canonical object.

Remark 2.5 (Definition, tie-breaking, and layer assignment). The definition of κ^μ is fixed as stated: all empirical claims of this manuscript use it in exactly this form. (i) The carrier is the flux-layer signed crease measure D^2v^* (Definition 2.1 applied to $u = v^*$) of the *three-stage lexicographic* optimal flux (§7); the lexicographic tie-break (biomass, then ℓ_1 flux, then seeded deterministic weights) is part of the metric’s definition, not an implementation detail — the flux layer is selection-dependent by construction, and the tie-break is declared rather than eliminated. (ii) The per-reaction mass is the unsigned total variation $\frac{1}{T} \sum_e \|A_e\|_1$ along the parameter trajectory. (iii) The value layer $D^2\Phi$ enters only as a *derived diagnostic* (Theorem 2.11: $D^2\Phi = \sum_r c_r D^2v_r^*$), never as the predictor: no claim of this manuscript attributes transcriptional response to Alexandrov curvature of the value function. (iv) The layer correlated with transcriptomics is the codimension-one Hessian/crease layer of the flux map — not the Monge–Ampère atom layer (Proposition 2.13), which requires concavity and serves as the dual-face diagnostic. Robustness to the declared selection is measured, not assumed: the within-layer rank identity $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ (§5.3), the trajectory substitutions of §5.5 reproduce the association under the same declared selection on two further paths, and §5.6 holds the association stable ($r \in [+0.386, +0.396]$, partial constant

to three decimals) under four alternative stage-3 selection rules while the value layer stays exactly invariant.

2.3 The value-function layer (Alexandrov)

Parametric linear programming also produces a scalar object: the optimal value $\Phi(\theta)$, the maximal growth rate in the metabolic setting. The value function is canonical — it does not depend on which optimal flux the solver returns — and classical convex analysis attaches to it a curvature measure of its own, which this subsection develops.

Proposition 2.6 (Existence and tie-break-freeness). *Let $\Phi(\theta) = \max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine, feasible and bounded. Then Φ is concave (pointwise infimum of affine dual objectives over a θ -independent dual feasible set), and $-\Phi$ carries a symmetric matrix-valued Radon Hessian measure $D^2(-\Phi)$ (Alexandrov (Alexandrov, 1939)), independent of the optimal-flux selection rule. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it. For OR-containing rules the value function remains continuous piecewise affine and carries the signed crease measure of Proposition 2.12; only the Monge–Ampère layer requires the concave class.*

Proposition 2.7 (Two-layer structure of $D^2(-\Phi)$). *Let f be convex piecewise affine. (i) The singular part of D^2f is supported on the codimension-one facets, with density $(g^+ - g^-) \otimes n = [\partial_n f] n \otimes n$ per unit facet measure; it has no atoms: the mass in an ε -neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log-log slope 1.000). (ii) The Monge–Ampère measure $\det D^2f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\text{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under ε -shrinking).*

Proposition 2.8 (Dual optimal face identity). *At a codimension-two vertex θ_v of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = |\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 vs dual-face enumeration area 0.235000068 (16 direction LPs over the dual optimal face).

Remark 2.9 (Layer separation and PhPP positioning). The phenotype phase planes of Edwards–Ibarra–Palsson (Edwards et al., 2001; Ibarra et al., 2002) are the chamber complex of Φ ; their sector corners are the atoms of the Monge–Ampère layer. The empirical κ^μ of Definition 2.3 is a functional of the flux-map measure D^2v (Definition 2.1), which is tie-break-sensitive: under two lexicographic tie-breaks the value layer of a controlled follower-variable LP is invariant, while the flux layer’s second-difference mass moves by ten orders of magnitude (§4.1). The controlled relation between the value layer and the flux layer is a theorem (Theorem 2.11 and Corollary 3.9); the association results of §5 stand on the flux layer alone, which is where the evidence (§5.3, §5.4) actually lives.

2.4 The value–flux coupling

The two layers are not independent. The next results record the exact relation: the value measure is the objective’s contraction of the flux measure, and the flux-layer mass that the value layer cannot see is precisely the degenerate rerouting inside optimal faces.

Lemma 2.10 (Lex-selection regularity). *Let ℓ, u be continuous piecewise affine and let $v^*(\theta)$ be the three-stage lexicographic optimal flux (§7). Then v^* is well defined, continuous, and piecewise affine; its graph over each GPR cell is a finite polyhedral complex in (v, θ) -space (each lexicographic stage adjoins the graph of a piecewise-affine value function, which is polyhedral, and an injective polyhedral projection is piecewise affine). Measured: path segment residuals $\leq 8 \times 10^{-14}$ (§4.1); Φ piecewise affine at 4.2×10^{-13} (§4.4).*

Theorem 2.11 (Value–flux crease coupling). $\Phi = c^\top v^*$ pointwise; hence, as symmetric matrix-valued signed Radon measures on the parameter domain,

$$D^2\Phi = \sum_r c_r D^2v_r^*.$$

Along any piecewise-affine trajectory, at every crease time t_k : $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$, and $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$.

(i) (**Visibility dichotomy.**) *An event with unique optima on both sides is a transversal optimal-vertex switch and is necessarily objective-moving; objective-invisible events ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) are exactly the degenerate reroutings inside ≥ 1 -dimensional optimal faces (mask-type events). The tie-break sensitivity of the flux layer is concentrated in the invisible events; the c -contraction is tie-break-free because it annihilates the degenerate layer.*

(ii) (**Sparse-objective corollary.**) *If $c = \gamma e_{\text{bio}}$, then $D^2\Phi = \gamma D^2v_{\text{bio}}^*$: the value layer is the crease measure of the single biomass component, and per-gene attribution from the value layer is structurally impossible.*

In words: an optimal switch is visible to the value layer only when it moves growth; the objective-invisible events are zero-cost pathway substitutions — reroutings among flux patterns of equal growth — carrying large flux changes at unchanged value. Machine-verified: identity error 0.0 at all 12 events and all 4 clusters of the iML1515 cut ($\Phi = c^\top v^$ to 10^{-9} ; Fig 2); 11 of 12 events objective-invisible, with flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$; 150 random dense-objective LPs to 1.1×10^{-13} (5 events, all objective-moving — the dichotomy); the degenerate follower family 60/60 with invisible kinks at identity error 2.0×10^{-13} ; two-dimensional mixed second differences at crease-straddling boxes to 6.4×10^{-16} .*

Proposition 2.12 (Semiconvexity collapse; signed layer for mixed GPRs). *A continuous piecewise-affine function is semiconvex (resp. semiconcave) iff it is convex (resp. concave) — semiconvexity is not an available weakening for LP value functions (measured: the required constant blows up like $1/(2h)$; $\lambda \cdot h_{\max} = 0.500$ numerically across $\lambda = 1 \dots 10^6$, the analytic law $h_{\max} = 1/(2\lambda)$ — equivalently $\lambda h_{\max} = 1/2$ — holding at every scale, with the numerical grid flooring above $\lambda \approx 10^7$). Nevertheless every continuous piecewise-affine Φ , for any GPR structure, carries a finite signed symmetric Radon Hessian measure on its crease skeleton with per-crease sign flags: for the OR-plus-cap value function $\min(\max(\theta_1, \theta_2), K)$, the diagonal crease density is positive semidefinite (eigenvalues $\{0, \sqrt{2}\}$) and the cap crease negative semidefinite ($\{-1, 0\}$). Concavity is required only by the Monge–Ampère layer and the dual-face identity; the signed Hessian layer applies to all genes. Under multiplicative activity scaling with $\ell \leq 0 \leq u$, $\Phi = \Psi(a(\theta))$ with Ψ concave nondecreasing: AND-only (min-tree) GPRs retain concavity, top-level OR nodes can break it, and disaggregating isozymes into separate reactions restores affine bounds while replacing max-semantics by sum-semantics (capacity 1 vs 2 at full expression) — a different biological model, not a regularization.*

Proposition 2.13 (Monge–Ampère atoms are determinants, not products). *For convex piecewise-affine f on $\mathbb{R}^2$ at a generic vertex with incident gradients n_1, n_2, n_3 and codimension-one gradient jumps j_i : the atom equals $\frac{1}{2}|\det(j_1, j_2)| = |\text{conv}\{n_i\}| = \frac{1}{2}|j_1||j_2|\sin\angle$; the product $|j_1||j_2|$ overestimates by $2/\sin\angle \geq 2$, with equality iff the jumps are orthogonal (measured on 400 random*

210 *max-of-affine vertices: determinant = fan area to 8.9×10^{-16} ; product/atom median 3.296, mini-*
 211 *mum exactly 2.000 (the orthogonal edge case; 2.000031 over random vertices); the $2/\sin \angle$ law to*
 212 *2.5×10^{-12} ; in p dimensions, atom = $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans).*

213 *Remark 2.14* (Sign and normalization conventions). Definition 2.1 applied to $u = -\Phi$ is the
 214 *signed crease measure* of the value function: per facet F , the density $A_F = [\nabla(-\Phi)]_F \otimes n_F |F|$, a
 215 symmetric matrix whose eigenvalue signs flag the crease’s convexity/concavity (Proposition 2.12).
 216 This is the explicit extension of Alexandrov’s setting to arbitrary GPR structure: within the
 217 concave class $-\Phi$ is convex, every $A_F \succeq 0$, and the measure coincides with the Alexandrov
 218 Hessian measure; outside it, the same definition produces the signed layer. All masses in this
 219 manuscript are unsigned total variations (entrywise 1-norm for matrix atoms, trajectory sums for
 220 κ^μ); Monge–Ampère atoms are unsigned $\frac{1}{2}|\det(j_1, j_2)|$ equal to the normal-fan volume, normalized
 221 to $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans in dimension p (Proposition 2.13). Signed determinants
 222 appear only in the concave class, where they are nonnegative.

223 *Remark 2.15* (The exact value–flux relation). Theorem 2.11 is the exact form of the value–flux
 224 layer relation: the empirical κ^μ is the per-reaction total variation of the flux-layer crease measure
 225 whose objective contraction is $D^2\Phi$ — the metric is the primal layer of the canonical object,
 226 justified by identity rather than by rank agreement.

227 2.5 The categorical reading, in brief

228 The definitions and results of §2 are measure-theoretic, and the empirical claims rest on them
 229 alone. The categorical reading summarized in this subsection motivates the active-set curvature
 230 interpretation and is developed in full — constructions, machine verifications, and proofs, with
 231 the proof status marked per result (complete for the gluing formula, the transport law, the com-
 232 position and contraction layers, and the terminal-coalgebra theorem; machine-verified numerics;
 233 citation-level stratified descent; proof-sketch status for the homotopy-type-theoretic extension) —
 234 in a companion theory paper (Abae, 2026). None of the empirical results of this manuscript de-
 235 pends on it. For orientation: a *connection* is the rule for comparing optimal solutions at nearby
 236 parameter values, and its *holonomy* is the drift accumulated when a closed loop of parameter
 237 changes fails to compose to the identity — the geometric form of non-commutativity.

238 *Strata as a stratified base.* The lexicographic pFBA engine (§7.1) partitions parameter space
 239 into chambers on which the optimizer is affine; the codimension-one facets where the active set
 240 switches are exactly the facets that carry the atomic measure μ of Definition 2.1. In the categorical
 241 reading, this partition is a stratified base and the optimizer organizes the policy fiber stratum by
 242 stratum, as a stratified connection in the companion’s 2-category **StCon**(B): connection 1-forms
 243 on each stratum, glued along the switching facets by a boundary transition satisfying a gauge
 244 matching condition, with a gluing theorem (proved there by reduction to stack-descent theory,
 245 and with its piecewise-holonomy consequences machine-verified in five loop geometries and three
 246 structure groups).

247 *The active-set bridge.* Two results of the companion connect this geometry to viability (Abae,
 248 2026). First, on each constant-active-set stratum, a Fisher-minimal transport law defines the
 249 connection in closed form as a constrained projection with the Fisher–Rao metric; its curvature is
 250 the smooth avatar of the per-facet density that μ concentrates on. Second, a small-loop viability–
 251 holonomy theorem bounds the endpoint erosion of a viability margin around a small parameter
 252 loop by the *viability-weighted curvature*: the worst case, over unit-area parameter bivectors and
 253 active viability margins, of the positive part of the margin’s change along the connection curvature,
 254 normalized by the margin itself. The per-pair form of κ^μ is modeled on that object — the atomic
 255 crease measure is the measure-theoretic analogue of the curvature 2-form, and the value–flux
 256 coupling (Theorem 2.11) supplies the analogue of the survival covectors; the precise discretization
 257 correspondence (a domination or ratio-form identity between the two) is open.

258 *Regime delineation.* The two holonomy-scaling regimes of Proposition 3.5 are the measure-
 259 level image of the companion’s two loop regimes: the smooth arm (slope 2.000) is where the

companion's quadratic small-loop law $\kappa(a) = a^2$ and its Fisher-minimal transport law live; the piecewise-affine arm (slope 1.00) is the companion's wall-crossing regime, where a loop crossing a switching facet accumulates a boundary contribution at $O(\varepsilon)$ per crossing (the companion's pairs-crossing theorem).

Division of labor. The present paper states only this brief reading; the companion theory paper develops the SAVGS object, the gluing theorem and piecewise holonomy formula, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, and the homotopy-type-theoretic extension, with proofs and extensions.

3 The refinement–resolution bridge

Section 2 treated the discrete measure exactly. This section asks the refinement question: as a triangulation of parameter space is refined, does the discrete measure converge to the smooth curvature of a smooth response map? The answer splits — weakly and cell-wise yes, in total variation no — and a resolution parameter σ separates the two regimes. Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \text{diam}(\tau) \rightarrow 0$, and u_h is the continuous piecewise-affine interpolant of u on $\mathcal{T}_h$.

Theorem 3.1 (Refinement–resolution bridge). *Let $\mu_h = D^2 u_h$ (Definition 2.1) and $\mu = D^2 u \, \text{dvol}$. Then:*

- (i) (**Atoms / support.**) μ_h is a finite matrix-valued Radon measure concentrated on the $(p-1)$ -skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.
- (ii) (**Weak convergence.**) $\mu_h \Rightarrow \mu$ weakly as matrix-valued measures: for every $\varphi \in C_c^\infty(\Omega)$, $\langle \mu_h, \varphi \rangle \rightarrow \langle \mu, \varphi \rangle$. Moreover $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$ in the Kantorovich–Rubinstein (flat) norm.
- (iii) (**Total variation: the exact statement.**) $\sup_h \|\mu_h\|_{\text{TV}} \leq C |u|_{C^2} |\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$; but $\|\mu_h\|_{\text{TV}} \rightarrow \|\mu\|_{\text{TV}}$ is false in general — see Remark 3.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of κ^μ .
- (iv) (**No-mass-loss, correct form: L^1 reconstruction.**) On structured complexes, the per-cell reconstructed density $\rho_h(\text{cell}) = |\text{cell}|^{-1} \sum_{F \rightarrow \text{cell}} A_F$ converges to $D^2 u$ strongly in L^1 with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \rightarrow \|\mu\|_{\text{TV}}$. On unstructured complexes the unweighted three-facet patch is not the dual cell; the general statement requires the discreteness (dual-diamond) pairing.
- (v) (**Resolution window.**) For the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$: at fixed h , the limit $\sigma \rightarrow 0$ is the atomic measure; the smooth object exists only as a family member at matched resolution through the window $h \ll \sigma \ll L_{\text{var}}$, where L_{var} is the length scale on which the smooth curvature density itself varies (measured in §4.3).

All five parts are proved in Appendix B, the reconstruction part (iv) in Appendix C.

Remark 3.2 (Exact counterexample calculus). On the right-triangle mesh of $[0, 1]^2$ with Hessian $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the per-cell atoms are $A_{\text{diag}} = bh^2 \mathbf{1}\mathbf{1}^\top$, $A_{\text{vert}} = (a-b)h^2 e_x \otimes e_x$, $A_{\text{horiz}} = (c-b)h^2 e_y \otimes e_y$; their sum is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a-b| + |c-b| + 4|b|$ against the target $|a| + |c| + 2|b|$. Hence for $u = xy$ the ratio is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.6–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

Proposition 3.3 (Two-layer separation and the sec law). *Let $K_h = \sum_v \text{defect}(v) \delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then K_h converges weakly to the Gauss-map area density*

$$\frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}} \text{dvol} = K \sqrt{1 + |\nabla u|^2} \text{dvol}, \quad K = \frac{\det D^2 u}{(1 + |\nabla u|^2)^2},$$

i.e. curvature per projected parameter area, not the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1 + |\nabla u|^2}$ (the sec law): machine-verified to 0.0% excess across 13 probes spanning tilts 0.11–2.83 and three tilt directions. Recovering intrinsic curvature requires the graph area correction.

Remark 3.4 (Consequences). (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 3.3; (iii) defects of a *fixed* complex (as measured in the regime-dial census, §4.3) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 3.3, which concerns refinement limits.

Proposition 3.5 (Regime dichotomy, in closed form). *Let $B(\varepsilon)$ be the curvature mass captured by a loop of radius ε . For a C^2 map, $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the regime-dial design (§4.3): 2.000 (smooth) and 1.00 (parametric FBA, 76-pair perturbation scan).*

3.1 Bridge to parametric FBA

The bridge of Theorem 3.1 concerns interpolants of a smooth map. The natural parametric-FBA analogue replaces interpolation by sequences of LP value functions, and is stated as a conjecture.

Conjecture 3.6 (mpLP refinement bridge). *Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; θ in the right-hand side; nested outer approximations) converging uniformly to a C^2 limit f , with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2 \Phi_n$ converges weakly to $D^2 f$ dvol provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.*

Remark 3.7 (Status). The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \rightarrow 0.0039$, mass ratio $0.939 \rightarrow 1.0009$ across $n = 4 \rightarrow 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

Remark 3.8 (Existence vs. resolution). Proposition 2.6 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem 3.1 governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

Corollary 3.9 (Decoupling; resolves the value–flux layer relation). *Theorem 2.11 resolves the layer relation exactly: value events are a subset of flux events, never the converse, and the flux events invisible to the value layer are the degenerate reroutings (mask-type events — rerouting without growth loss, the Kochanowski pattern (Kochanowski et al., 2013)). On the iML1515 cut: 12 flux events vs 1 value atom, 11 objective-invisible; on the reference trajectory (§5.4) the value layer carries no network events at all. The association of κ^μ with transcriptional response is therefore a flux-layer property by structural necessity, not merely by empirical choice.*

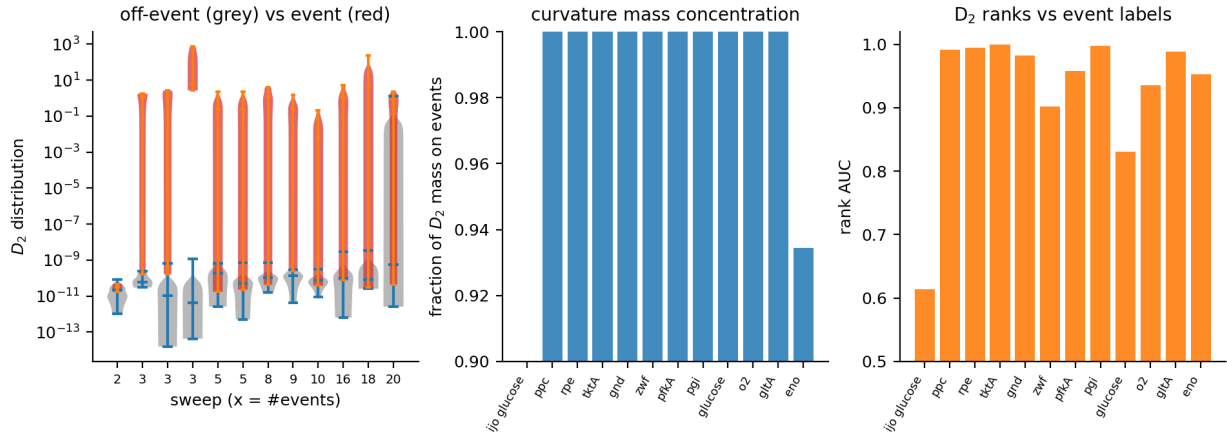


Fig 1. Active-set curvature verification (iML1515). Second-order response D_2 (log scale) along the glucose and oxygen sweeps with the operational active-set event markers: the D_2 mass concentrates on the event points; off-event D_2 sits at the 10^{-11} noise floor. The engine is the three-stage lexicographic pFBA solver of §7; all sweeps are cold-started and deterministic.

4 Computational validation

The geometric claims of Sections 2 and 3 are operational: they refer to the active set of a solved linear program, and can therefore be tested directly on genome-scale models. Each subsection below reports one such test.

4.1 Active-set curvature verification

We first verify the support claim: second-order response mass should concentrate on active-set switches. Thirteen parameter sweeps (glucose and oxygen uptake; capacity knockdowns of ten genes: *pgi*, *zwf*, *tktA*, *pfkA*, *eno*, *gltA*, *aceA*, *ppc*, *gnd*, *rpe*) on *E. coli* iML1515, solved by a three-stage lexicographic pFBA engine (deterministic optimum; fixed seeded tie-break weights resolving the vertex degeneracy of plain pFBA), with an iJO1366 glucose replication and an *aceA* unused-pathway negative control. Result: 93.4–100.0% of second-order response mass concentrates on operational active-set switches in every sweep that crosses critical-region boundaries (ten of the eleven crossing sweeps at mass exactly 1.000000; mass 0.934–1.0; AUC 0.83–1.00; Mann–Whitney $p \leq 2 \times 10^{-3}$, with seven of eleven crossing sweeps at $p \leq 10^{-4}$; Fig 1); between events the paths are piecewise affine to solver precision (event-free segment residuals $\leq 1.2 \times 10^{-10}$, with 8×10^{-14} on the glucose sweep; one 19-point segment of the *eno* sweep carries 4.1×10^{-3} — a sub-threshold kink invisible to the operational event proxy, which is also why that sweep’s event mass is 0.934 rather than 1.0); negative controls behave as theory dictates (unused-pathway knockdown: zero events, zero curvature; single-critical-region sweep: noise-level $D^2 \sim 10^{-11}$).

4.2 Double-knockout epistasis and path dependence

Epistasis — the interaction between two knockouts beyond the sum of their individual effects — provides a combinatorial test of the active-set reading: if the measure is a genuine geometry, pairs of perturbations should interact through it. 1,516 single knockouts and 2,779 double-knockout pairs across five panels. Epistasis $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$: all 40 synthetic-lethal pairs are isozyme redundancies (*tktA*/*tktB*, *acnA*/*acnB*, *metE*/*metH*, ...) with pure-emergence epistasis; $|\varepsilon_{ij}|$ aligns with active-set footprint overlap (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$); sequential (L_1 -MOMA (Segrè et al., 2002)) knockouts open nonzero commutators in 25% of active pairs; closed genotype loops fail to return to the initial state in 66% of pairs (phenotypic memory; median holonomy ~ 110).

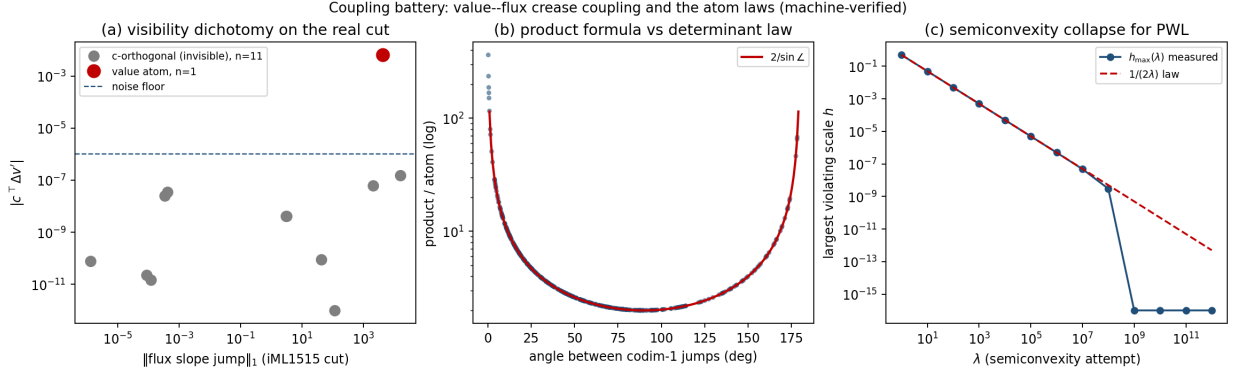


Fig 2. Machine verification of the value–flux crease coupling (Theorem 2.11). The controlled follower-variable LPs: the value function Φ stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips; the coupling identity $\Delta\Phi' = c^\top \Delta v'$ holds to 0.0 on the toys, 1.1×10^{-13} on 150 random dense-objective LPs, and 2.0×10^{-13} on the 60/60 degenerate follower family; on the iML1515 cut the identity error is 0.0 at all 12 events and all 4 clusters.

4.3 Regime dial

The discrete and smooth regimes of Section 3 can be separated empirically by setting the loop size ε against the smoothing scale σ and dialing one against the other. The smoothing identity $D^2(v * \phi_\sigma) = (D^2 v) * \phi_\sigma$ holds to kernel self-test error 1.2×10^{-6} ; measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$ discrete regime. The scaling slope is 1.00 on the interacting stratum; the regime-dial census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal ± 1884.6).

4.4 The value function as canonical carrier

The tie-break-freeness of the value layer (Proposition 2.6) is directly testable: solve the same parametric problem under two different tie-break conventions and compare layers. $\Phi = c_{\text{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at 4.2×10^{-13} on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio 1.7×10^{-6}). Danskin: finite-difference shadow prices match envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 2.6), the flux-strain layer is not (Corollary 3.9). The coupling identity of Theorem 2.11 is machine-verified on this same cut: $\Delta\Phi' = c^\top \Delta v'$ with error 0.0 at all 12 events and all 4 clusters, $\Phi = c^\top v^*$ to 10^{-9} , and the 11 invisible events carry flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$ — the decoupling is the objective contraction, not a loss of network information.

5 Empirical association and protein-layer buffering

This section takes the metric to data: which genes does the network itself flag as rerouting-sensitive, and do those genes respond when the environment actually tightens? The answer comes in layers — a robust transcriptional association, an explicit negative result at the protein layer, and robustness checks across trajectories, tie-break conventions, and panels.

5.1 The gene panel and GPR classes

Reaction-based sampling on the reference physiology (glucose $5.0 \rightarrow 1.0$, oxygen $22 \rightarrow 5$; iJO1366): 438 of 2,583 reactions active (17.0%), 435 genes with at least one active reaction. GPR complexity classes over active reactions: single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34 — the classes that drive the concavity analysis of Proposition 2.12. Panel construction passed four verification checks: (i) GPR mapping taken from cobra’s authoritative `reaction.genes`, spot-checked against the raw rule strings on four reactions spanning all complexity classes, with set-equality of gene tokens and a bidirectional gene–reaction symmetry check on 50 sampled genes (0 failures over 120 gene–reaction pairs); (ii) definition consistency — the fresh re-derivation reproduces the deposited values exactly (per-gene maxima, the biomass-deficit maximum, and the unmasked/masked panel correlations to four decimals); (iii) distinctness — the panel shares exactly one b-number with the κ_V panel (*b2097*); (iv) non-zero variation — every panel gene has κ_V variation > 0 over T1–T8 (435/435). The geometric metric κ_V is weak at the panel level (r from -0.063 unmasked to $+0.084$ masked); the measure-theoretic κ^μ (§5.3) is the recalibrated predictor on the same panel.

5.2 Multi-condition support

Nine FBA conditions (baseline carbon decline; galactose, trehalose, glycerol-3-P, and proline-N carbon sources; anaerobic-nitrate, oxidative, and osmotic designs): active-reaction support 438–454 per condition, union 537 (20.8%); genes with nonzero κ_V expand $435 \rightarrow 525$ across conditions. The endogeneity check is clean (feed caps non-binding except the designed glycerol-3-P limitation; imposed-burden magnitudes flagged as not interpretable). The rerouting metric is well posed across carbon sources and stress designs: its support *expands* with condition diversity rather than migrating.

5.3 Transcriptional response under carbon depletion

On the deterministic lexicographic trajectory (the reference physiology, iJO1366), κ^μ (Definition 2.3) versus log-fold carbon-depletion response (M3D (Faith et al., 2008), $n = 424$ genes; Fig 3): Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$ (full panel, zeros included), partial $r = +0.269$ controlling reference level ($p = 1.8 \times 10^{-8}$), top-decile versus bottom-decile response 1.92 vs 0.89 (MWU 1.3×10^{-7}). Baseline reproduced to the digit ($r = +0.3739$, $n = 433$); metric invariance $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ — both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer, not a metric artifact and not a value-function shadow. Cross-platform replication with the declared metric: the carbon-switch arms replicate on the same platform (M3D microarray: glycerol $+0.195$, acetate $+0.166$, proline $+0.175$, switch-MAX $+0.206$; $n = 424$ genes, all $p < 6 \times 10^{-4}$) but dissociate across platforms — the PRECISE RNA-seq carbon-switch statistic (ten WT non-glucose carbon conditions) gives $r = -0.044$ (partial -0.088 , NS), with individual conditions weakly positive (glycerol $+0.126$, acetate $+0.130$, fructose $+0.099$) except galactose (-0.088). The association is specific to the carbon-exhaustion class and the platform-matched design; the cross-platform dissociation is reported as a negative result.

5.4 The layer decision

All value-layer attribution arms, under the declared protocol of §5.3 (iJO1366, the reference physiology, $n = 424$ genes): the flux layer κ^μ gives $r = +0.395$, partial $r = +0.269$ ($p = 1.8 \times 10^{-8}$); the c -attribution arm is supported on the biomass pseudo-reaction alone (0 of 433 genes; Theorem 2.11(ii)); the shadow-price-jump arm gives 51 genes at $r = +0.032$ (partial -0.013 , $p = 0.93$). The value trajectory has *zero* chamber crossings: the path stays in one chamber of Φ throughout — Φ is affine along it, with $d\mu/dq_{\text{glc}} = Y = 0.099544$ ($= |y_{\text{glc}}|$, the constant glucose

uptake shadow price; affine intercept -0.0124) and uptake shadow prices jumping by exactly zero — all four value kinks are anchor corners of the trajectory design, of magnitude $Y \Delta(dq_{\text{glc}}/dt)$, carrying no network structure. Value/flux strain mass ratio 1.45×10^{-3} ; the coupling identity holds at machine precision (error 0.0 at every kink). The value-gated flux strain coincides with κ^μ exactly (433/433 genes attain their maximum at a value-kink time), a property of the trajectory design (anchor rerouting dominates per-gene maxima), not evidence of value-layer content. The layer decision: the association lives on the flux layer, with Theorem 2.11 as the exact bridge to the canonical object.

5.5 Path robustness: what survives a trajectory change

Three trajectories through the same LP (same engine, seed, panel, statistics; only the path and its matched response vary; Fig 4): P0 glucose decline (reference-physiology anchors; the layer-decision control of §5.4), P1 oxygen limitation ($q_{\text{glc}} = 5$ fixed, q_{O_2} linear from 22 to 1), P2 acetate switch ($q_{O_2} = 22$ fixed; glucose $5 \rightarrow 0$ while acetate uptake ramps $0 \rightarrow 10$). Kinks are classified by the chamber test — the parameters are the uptake bounds, so $\nabla\Phi = (y_{\text{glc}}, y_{O_2}, y_{\text{ac}})$: a kink is a *design corner* iff it sits at a trajectory anchor and the uptake shadow prices are continuous across it (jump $\leq 10^{-9}$), and a *chamber crossing* iff they jump.

- (i) The one-chamber property is *path-dependent*: P0 has zero chamber crossings (the §5.4 control, reproduced digit-exactly), while P1 has three genuine crossings ($q_{O_2} \approx 9.3, 6.3, 2.9$; $|\Delta\Phi'|$ up to 0.68; y_{O_2} jumps $+0.032$ at the respiratory boundary) and P2 has four (two large, near the glucose–acetate handoff; two micro-slivers with dual jumps $\sim 10^{-4}$). The single-chamber structure of the reference trajectory is a property of the glucose-decline design, not a law.
- (ii) The layer decision survives the change: on every path the value-layer arms contribute nothing. The c -attribution arm is degenerate (0 of 433 genes — the sparse-objective corollary of Theorem 2.11 is path-free); the chamber-crossing shadow-price arm is null ($r = -0.17$, $n = 68$ on P1; $r = +0.032$, $n = 71$ on P2); value-gated flux strain is a subset of κ^μ (equal or weaker on every path, never stronger).
- (iii) The flux-layer association generalizes: against the matched oxygen response (M3D WT anaerobic vs aerobic contrast) P1 gives $r = +0.318$ (partial $+0.258$, $p = 6 \times 10^{-8}$, $n = 426$); against the matched acetate-switch response P2 gives $r = +0.223$ (partial $+0.160$, $p = 9 \times 10^{-4}$); and the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378 / +0.391$ — the per-gene ranking is largely trajectory-independent.
- (iv) Theorem 2.11 holds at every kink on all three paths (identity error 0; worst case 1.6×10^{-15}). Value/flux strain mass ratios: 1.5×10^{-3} (P0), 2.4×10^{-4} (P1), 5.5×10^{-4} (P2).

5.6 Tie-break robustness: the association as a property of the protocol class

The flux layer is selection-dependent by construction (Remark 2.5); the declared lexicographic tie-break is one member of a family. The tie-break experiment fixes the entire protocol of the primary association (iJO1366, reference-physiology anchors, $8\times$ refinement, the 433-gene panel, the M3D carbon-depletion response, the reference-level confound control) and varies *only* the stage-3 tie-break of the engine:

- **TB0** (declared): $w \sim U(0.5, 1.5)$, seed 20240901, $\min w^\top v$ — the declared metric (control: reproduces the primary association, $r = +0.3954$, digit-exactly);
- **TB1** (fresh seed): $w \sim U(0.5, 1.5)$, seed 20240902 — same family, independent draw;
- **TB2** (family swap): $w \sim \text{LogN}(0, 1)$ clipped to $[0.05, 20]$, seed 20240903 — a different distribution family with a roughly 130-fold wider dynamic range (support ratio 400:1 versus 3:1);

Table 1. Tie-break robustness of the association (primary protocol fixed; only the stage-3 selection rule varies). The value layer (stage-1 biomass μ) and the pFBA layer (stage-2 ℓ_1, s_2) are invariant to 0.0 across all variants — the measured form of the tie-break-freeness of Proposition 2.6.

Variant	n_{nz}	r	partial r	$\rho_S(\kappa^\mu, \kappa_{\text{TB0}}^\mu)$	genes changed
TB0 declared	424	+0.3954	+0.269	—	—
TB1 fresh seed	426	+0.3861	+0.270	0.99998	91
TB2 LogN family	424	+0.3954	+0.269	0.99999	90
TB3 $ v $ rule	425	+0.3959	+0.269	0.99897	109
TB4 max $w^\top v$	424	+0.3954	+0.269	0.99998	117

- **TB3** (rule swap): stage-3 objective on the split variables, $\min \sum_r w_r(f_r + r_r)$ — weighted *absolute*-flux selection with TB0’s weights;
- **TB4** (adversarial): $\max w^\top v$ — the far end of the stage-2-pinned optimal face, with TB0’s weights.

Three findings (Table 1, Fig 5). (i) *The association is stable*: across all five rules $r \in [+0.386, +0.396]$ (maximum deviation 0.009; all $p \leq 1.4 \times 10^{-16}$) and the partial correlation is constant to three decimals (+0.269–+0.270). (ii) *The tie-break genuinely binds*: the stage-2-pinned optimal face is multi-vertex, and the alternative rules select different flux vectors at all 57 grid points ($\max \|v_k - v_0\|_\infty$ up to 5.0), changing per-gene κ^μ for 90–117 of 433 genes — yet the per-gene ranking is essentially unchanged ($\rho_S \geq 0.99897$), and the changed values are small on the κ^μ scale (max absolute change 9.3×10^{-5}); the TB1 deviation of 0.009 traces to two boundary genes entering the nonzero support ($n : 424 \rightarrow 426$), not to a reshaping of the metric. (iii) *The value layer is exactly invariant*: stage-1 biomass μ and stage-2 ℓ_1 mass agree across variants to 0.0 — Proposition 2.6 (tie-break-freeness of the value layer) measured on the full trajectory. Together with Corollary 3.9 this completes the robustness picture: the declared selection is a convention, the value layer does not see it, and the flux-layer association is a property of the lexicographic protocol class, not of one rule.

5.7 Event-measure stabilization (Glivenko–Cantelli type)

The refinement-stabilization question (*do the empirical event point measures of the lex-pFBA map converge as the sampled family grows?*) has a near-term statistical form, executed here on three data sources: random straight cuts through the $(q_{\text{glc}}, q_{\text{O}_2})$ signature plane (iML1515, 34×34 census, 350 boundary edges; a pool of 4,000 random cuts yielding 25,107 cut events), the thirteen parameter sweeps of §4.1 (2 affine, 11 event-bearing), and the panel protocol of this paper (gene panel and trajectory). The metric is the bounded-Lipschitz distance (domains normalized to unit diameter, where it equals W_1 ; exact in one dimension, transport LP in two); each arm carries a permutation null (event locations replaced by uniforms, panel structure kept) and the one-dimensional arms carry the classical Glivenko–Cantelli asymptotic constant $C = \int \sqrt{F(1-F)} dx$ of del Barrio–Giné–Uzet computed from the population (Fig 6).

- (i) *Random panels stabilize at the GC rate*. Pooled cut-event locations on the cut coordinate ($d=1$): measured tail slope -0.492 versus null -0.506 , measured/null ratio 1.24 at the largest panel, and the iid prediction $C/\sqrt{n}$ matches the measured curve to 2.5% at the largest panel (mid-range deviations as large as 8.2%); the two-dimensional pooled locations (transport LP) decay consistently with the $d = 2$ rate up to the reference-noise floor. Sweep panels (the thirteen sweep families) stabilize with a heterogeneity penalty that falls from $\approx 1.8\times$ the uniform null at small panels through $\approx 1.5\times$ at $k = 6$ to $\approx 0.5\times$ at $k = 12$, and a tail faster than iid-GC — consistent with the finite-population correction $\sqrt{(13-k)/12}$ of sampling

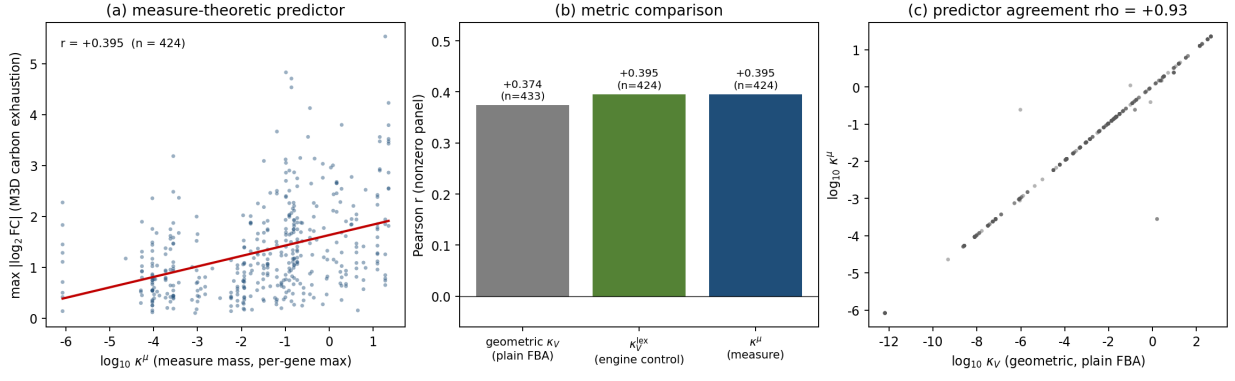


Fig 3. The primary association. (a) Per-gene $\log_{10} \kappa^\mu$ (flux-layer measure mass, per-gene max over the gene’s reactions) versus M3D carbon-depletion response ($\max |\log_2 \text{FC}|$, four stationary contrasts, $n = 424$ nonzero genes); (b) metric comparison: the geometric time-course κ_V (reference physiology, plain-FBA), the engine control κ_V^{lex} , and the measure-theoretic κ^μ on the same trajectory; (c) predictor agreement between κ^μ and the plain-FBA κ_V .

without replacement from thirteen heterogeneous sweeps (measured null-relative ratio 0.49 at $k = 12$, against the pure FPC value 0.29; the residual is the sweep heterogeneity). The gene panel: the κ distribution stabilizes at the GC rate (measured effective constant ≈ 1.6 –2.1 across panel sizes (dipping at $m = 128$), against the population Glivenko–Cantelli constant $C = 2.49$ on the $\log_{10} \kappa^\mu$ axis), and the association itself stabilizes at $r \approx +0.39$ with the Fisher standard error $1/\sqrt{m-3}$.

- (ii) *Designed panels stabilize exactly, not statistically.* The trajectory event point measure is a *four-atom* object: the four design corners of the physiology anchors carry the entire flux-layer mass (288.77, the primary association’s flux-layer mass to the digit) — the one-chamber structure of §5.4 in its event form. Anchor-preserving thinning reproduces it *exactly* at every panel size $m \geq 8$ (shape distance 0 to machine precision, $\leq 7 \times 10^{-13}$; total-mass error $\leq 2 \times 10^{-12}$; value-kink census $\equiv 4$); uniform random thinning decays by corner censoring (shape distance $0.12 \rightarrow 0.005$ by $m = 43$, mass error $26\% \rightarrow 0.13\%$ by $m = 29$, machine-zero $(\leq 3 \times 10^{-12})$ from $m = 43$, and exactly 0 at the full panel $m = 57$) — a resolution effect in the sense of Theorem 3.1(iv), not statistical averaging.

Interpretation: the mean-field intuition is confirmed in its statistical form — the empirical event measures over random panels (cuts, sweeps, genes) are Glivenko–Cantelli objects — and simultaneously sharpened: stabilization has two regimes, statistical for random panels and structural (exact, design-immediate) for designed panels. The deterministic refinement-sequence form of the question (refinement along explicit model-family sequences) remains open.

5.8 Platform-by-class disambiguation (2×2)

Is the association a carbon-depletion artifact, or a platform artifact? Four cells (two platforms $\times$ two perturbation classes), executed with the geometric κ_V metric (rank-equivalent to κ^μ at $\rho = 0.99998$): carbon switch on microarray (M3D glycerol/acetate/proline vs glucose) switch-MAX $r = +0.196$; stress on microarray (heat shock, acid shock, ciprofloxacin) MAX $r = +0.298$; carbohydrate-responsive transcript levels: carbon starvation on RNA-seq (GSE64021 (Barrett et al., 2013) time course) MAX $r = +0.191$; heat shock on RNA-seq MAX $r = +0.345$ ($n = 241$ –433; permutation $p \leq 0.003$). Signed responses flip sign (stress arms -0.22 to -0.32): the metric tracks *rerouting magnitude*, not direction, on both platforms and both classes — the association is neither platform-specific nor carbon-class-specific.

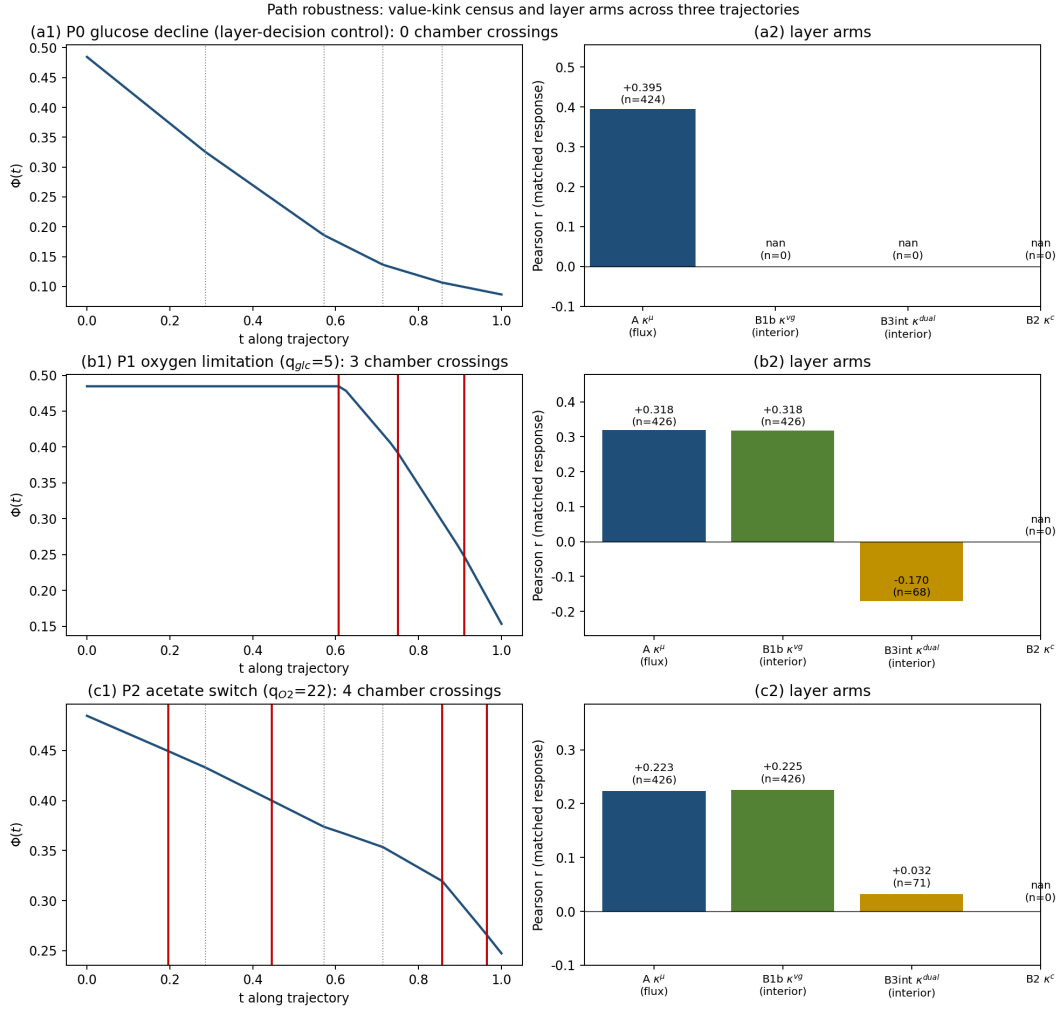


Fig 4. Path robustness. Three trajectories through the same LP (P0 glucose decline, P1 oxygen limitation, P2 acetate switch) with matched M3D responses: the one-chamber property is path-dependent (P1: 3 chamber crossings; P2: 4), while the layer decision and the flux-layer association survive on every path; the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378/+0.391$.

5.9 Protein abundance versus protein change

Two dissociations at the protein layer (PaxDb integrated abundance (Wang et al., 2015); quantitative proteomics of GSE64021 (Barrett et al., 2013); $n = 169-429$): (i) abundance *level* associates (PaxDb baseline $r = +0.334$) — high-rerouting genes are abundant proteins; (ii) abundance *change* does not (protein log-fold change $r = +0.008$ to $+0.032$; transcript–protein fold-change coupling $r = +0.020$, $n = 799$). The transcriptional association stops at translation, consistent with buffering.

5.10 Protein-layer null

Condition-dependent quantitative proteomics (Schmidt et al. (2016); 22 conditions, triplicates, $n = 366$): protein-level correlation $r = -0.083$ against transcript $+0.419$ on the same genes. The transcript-level association does not propagate to the protein layer; buffering at translation is the consistent model, positioning Kochanowski et al. (2013) as prior art for the protein-layer null.

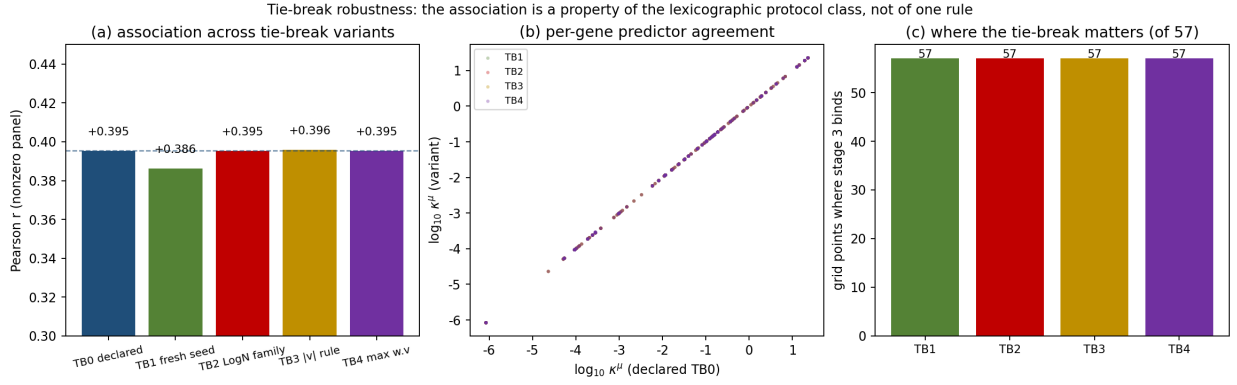


Fig 5. Tie-break robustness. (a) The association across the five stage-3 selection rules (declared plus four variants); (b) per-gene predictor agreement between the declared rule and each variant; (c) the number of grid points at which the tie-break binds (of 57). The value layer is invariant to 0.0 across all variants.

6 Discussion: one measure, several resolutions

The sensitivity objects of the framework are one measure at different resolutions: the atomic measure of Definition 2.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories (κ^μ). The smooth member exists only through the window $h \ll \sigma \ll L_{\text{var}}$; the $\sigma \rightarrow 0$ limit is the atomic object. Open items: Conjecture 3.6 (general complex stability) and second differences on measured time courses (under the (ε, σ) design law). The statistical form of the stabilization question is settled here (§5.7): random panels are Glivenko–Cantelli objects and designed panels reproduce the event measure exactly; the deterministic refinement-sequence form (refinement along explicit model-family sequences) remains open. The exact contraction identity $D^2\Phi = \sum_r c_r D^2v_r^*$ of Theorem 2.11 grounds the association’s flux-layer footing, complementing the measured metric invariance of §5.3. The empirical association reflects the separation: gene expression changes correlate with the flux-layer metric κ^μ ($r = +0.395$) but show no association with value-layer derivatives (shadow-price arm $r = +0.032$, $p = 0.93$, $n = 51$); transcriptional response tracks rerouting, and rerouting is a flux-layer quantity.

Limitations. The study is correlational by design, and its scope is a single organism. The active sets used throughout are operational (obtained from solved LPs) rather than combinatorial. The lexicographic tie-breaking is an engine convention: the value-function carrier is immune to it, the flux layer declares it (Remark 2.5), and the association is stable across all five selection rules tested (declared plus four variants, §5.6). The one-chamber structure of the reference path is condition-specific (§5.5), while the layer decision itself is path-robust. The cross-platform carbon-switch dissociation (PRECISE arm, §5.3) bounds the generality of the association. The active-reaction census of the reference physiology (438/2,583 and the GPR class counts) is plain-FBA and solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel (435) is stable (Appendix A). The sweep arm of the stabilization experiment is panel-limited (thirteen sweep families, two of them affine — the finite-population correction is part of the reported result), and the two-dimensional arm is floor-limited at large panels by the transport-LP atom cap.

The companion theory paper. The categorical and homotopy-theoretic reading behind the active-set curvature interpretation — the 2-category $\mathbf{StCon}(B)$ of stratified connections with its gluing theorem, the Fisher-minimal transport law on constant-active-set strata, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, the viability-kernel reading of its dynamical closure test (a finite-time, feedback-certifying probe of kernel and capture-basin membership, with the Poincaré/averaging analysis

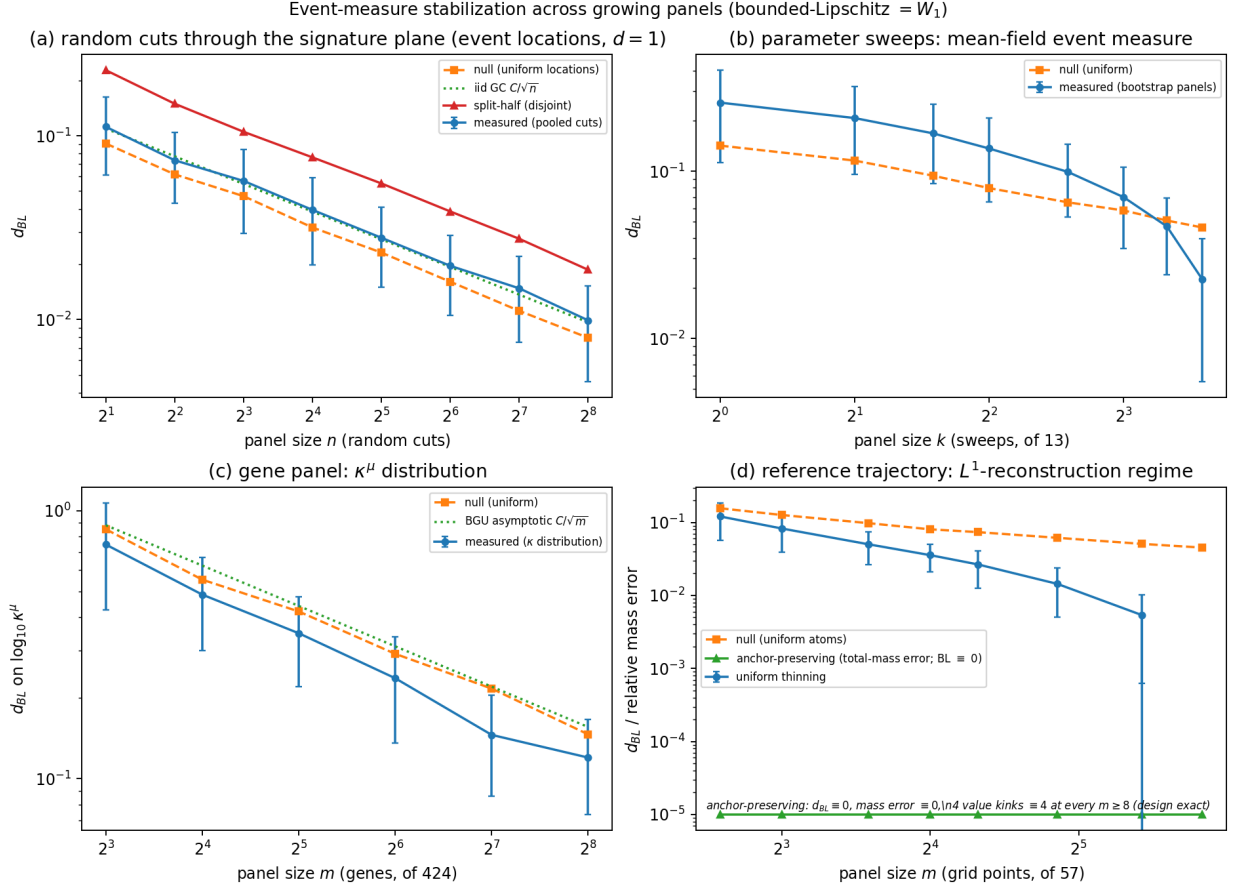


Fig 6. Event-measure stabilization. (a) Random-cut panels: pooled event-location distance decays at the Glivenko–Cantelli rate (measured tail slope -0.492 ; null -0.506 ; iid prediction $C/\sqrt{n}$ to 2.5% at the largest panel, mid-range deviations as large as 8.2%); (b) Parameter-sweep panels: stabilization with a finite-population-corrected, heterogeneity-penalized tail (null-relative ratio 0.49 at $k = 12$ against the pure FPC 0.29); (c) Gene panel: the κ distribution at the GC rate (effective constant ≈ 1.6 – 2.1 , dipping at $m = 128$, vs. the population constant $C = 2.49$); (d) Reference trajectory: the L^1 -reconstruction regime — random thinning decays by corner censoring, anchor-preserving thinning is exact ($d_{BL} \equiv 0$) from $m \geq 8$.

of its pathwise levels), and the homotopy-type-theoretic extension — is developed in full, as a standalone theory paper with its own proofs, machine verifications, and per-result proof-status marking (the homotopy-type-theoretic extension is proof-sketch-marked there; the stratified gluing is descent-by-citation with construction-level verification), in a companion manuscript (Abaee, 2026). The division of labor is deliberate: this paper is the application — the measure-theoretic curvature of parametric FBA, the metric κ^μ , and the genome-scale empirical association with its layer decision — and carries only a brief adapted statement of the load-bearing definitions (§2.5); the companion carries the framework. Neither manuscript depends on the other’s claims.

Scope of the contribution. The mathematics is classical: parametric linear programming (chamber complexes, active sets, Danskin duality (Danskin, 1967)), Alexandrov’s theorem (Alexandrov, 1939), and distributional second derivatives of piecewise-affine maps; Theorem 2.11 is an elementary identity. The contribution is the systematic *application* of this machinery to flux-balance models (Varma and Palsson, 1994; Orth et al., 2010; Lewis et al., 2010) — the two-layer measure structure of the value/flux pair, the exact coupling identity, the signed extension to arbitrary GPR structure — together with the empirical association and its layer decision. No new measure-theoretic theorem is claimed; the nearest prior art (phenotype phase planes (Edwards et al., 2001;

Ibarra et al., 2002), mpLP value-function theory (Borrelli et al., 2003), discrete Monge–Ampère (Gutiérrez, 2001), Regge-type defect measures (Regge, 1961; Cheeger et al., 1984)) is cited where it applies.

7 Methods

7.1 The lexicographic pFBA engine

All deterministic trajectories are computed by a purpose-built three-stage lexicographic solver on the variable split $x = (v, f, r)$ with $v = f - r$, $f, r \geq 0$, stoichiometry $Sv = 0$, and flux bounds $\ell(\theta) \leq v \leq u(\theta)$ entering affinely through the uptake bounds. The stages are strict lexicographic optimizations, each pinned at the previous stage’s optimum:

$$\begin{aligned} (1) \quad & \mu^* = \max c_{\text{bio}}^\top v && \text{(growth)} \\ (2) \quad & s^2 = \min \sum_r (f_r + r_r) \quad \text{s.t. } v_{\text{bio}} \geq \mu^*(1 - \varepsilon_\mu) && \text{(parsimony)} \\ (3) \quad & \min w^\top v \quad \text{s.t. } \sum_r (f_r + r_r) \leq s^2(1 + \varepsilon_s) && \text{(tie-break)} \end{aligned}$$

with $\varepsilon_\mu = \varepsilon_s = 10^{-9}$ and $w \sim U(0.5, 1.5)$ drawn once with seed 20240901 and fixed thereafter. Stage 3 exists because plain pFBA is vertex-degenerate on these models: repeated warm-started solves of the same LP flip 0.69 reactions on average between vertices of the optimal face. The LPs are solved with HiGHS (Huangfu and Hall, 2018) via `scipy.optimize.linprog`, cold-started and deterministic; a presolve-disabled retry handles the rare infeasibility mis-report. The lex optimum exists, is unique by construction, and is continuous piecewise affine in θ (Lemma 2.10); the tie-break is part of the metric’s definition (Remark 2.5), and its robustness is measured in §5.6. The active-reaction census of the reference physiology is the one plain-FBA computation retained (kept for comparability with the geometric metric κ_V); it is solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel is stable.

7.2 Models and data provenance

Models. *E. coli* iJO1366 (2,583 reactions) (Orth et al., 2011) for the association experiments of §5; iML1515 (Monk et al., 2017) for the computational batteries of §4 and the value–flux coupling verification cut. Genome-scale models are loaded from the local BiGG JSON copies with SHA-256 manifests.

M3D microarray compendium. The expression panels come from the M3D *E. coli* compendium v4 Build 6 (Faith et al., 2008, 2007): 4,297 gene probes $\times$ 907 arrays, $\log_2$ uniformly normalized. The carbon-depletion response is the maximum $|\log_2 \text{FC}|$ over the four stationary time points (135/330/480/720 min) against the MOPS glucose reference; the oxygen-limitation and acetate-switch responses use the M9 WT anaerobic versus aerobic and MOPS acetate versus glucose contrasts. The reference-level confound control is the mean log-phase WT MOPS glucose expression. The 117 MB compendium tarball is not committed (repository size); it is re-downloadable from the recorded URL with the committed SHA-256 (e73088f7...).

PRECISE. The cross-platform arm uses the ten WT non-glucose carbon conditions of the PRECISE compendium (Sastry et al., 2019) (RNA-seq, TPM). PRECISE has no carbon-depletion class; the dissociation is reported as a negative result with per-condition values (§5.3).

Proteomics. Condition-dependent quantitative proteomics from Schmidt et al. (2016) (22 conditions, triplicates); integrated protein abundance from PaxDb (Wang et al., 2015); the GSE64021 (Barrett et al., 2013) time course supplies the RNA-seq carbon-starvation and the quantitative proteomics arms (§§5.8–5.9).

7.3 Panel construction and GPR association

The gene panel is built by reaction-based sampling: solve the lexicographic trajectory on the reference physiology (glucose $5.0 \rightarrow 1.0$, oxygen $22 \rightarrow 5$; eight anchors, $8 \times$ per-segment trajectory refinement), take the 438 reactions active ($|v| > 10^{-9}$ at any grid point) on the plain-FBA census of the reference physiology, classify each active reaction’s GPR rule (single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34), and assign each gene the maximum over its associated reactions. The expression panel consists of the 433 genes of this set carrying M3D response data; the primary correlation uses the 424 genes with nonzero κ^μ . The near-colliding gene/reaction counts (424/433/435/438/440/525/537) are disambiguated in Appendix A.

7.4 Statistical protocols

The predictor is $\log_{10}$ of the per-gene κ^μ (its distribution is log-linear over five decades); the response is $\max |\log_2 \text{FC}|$ of the matched contrast. Reported statistics: Pearson r with analytic p , Spearman ρ_S (raw predictor ranks, full panel and nonzero subset as labeled), partial r controlling the M3D reference level by linear residualization of both variables, permutation p (10^5 label permutations, Monte Carlo), bootstrap 95% CIs (10^4 resamples), and top-vs-bottom decile contrasts (one-sided Mann–Whitney U). All tests are two-sided except the decile contrast. The PRECISE per-condition arms use the same protocol with TPM log-fold changes. Every manuscript number traces to a deposited artifact file and is re-derived by the automated numeric verification described under Reproducibility (301 checks).

7.5 Reproducibility

All experiments deposit their result files and generating scripts in the public repository cited in the Data Availability statement. Independent-seed reproductions are recorded for the refinement prototype and the stress batteries. The layer-decision, path-robustness, and tie-break-robustness analyses, the cross-platform arm, and the event-measure stabilization re-run in minutes each on a single workstation. An automated numeric verification of 301 checks re-derives every manuscript number from the deposited artifacts and is re-runnable end-to-end.

A Disambiguation of near-colliding counts

Near-colliding counts in the manuscript, each traceable to a single deposited artifact file (Data Availability):

Value	Meaning (source)
2,583	iJO1366 reactions (model file; recomputed)
438	reactions active on the reference physiology, 17.0% (plain-FBA census, ± 1 solver sensitivity)
440	reactions with $D_2 > 10^{-8}$ along the reference trajectory (reaction-level event count, deposited event file)
433	panel genes with M3D response (panel file row count)
424	panel genes with nonzero κ^μ — the n of the primary correlation (deposited metric file)
426	nonzero genes on the P1/P2 paths (deposited path file)
435	genes with ≥ 1 active reaction, reference physiology (census file; recomputed exactly)
454	max active reactions across the nine multi-condition panels (deposited multi-condition results)
537	union of active reactions across conditions, 20.8% (deposited multi-condition results)
525	genes with nonzero κ across conditions (deposited multi-condition results)
4	trajectory atoms of the event measure — the four design corners carrying the full flux-layer mass 288.77; identical to the value-kink positions (deposited event files)
350	boundary edges of the $(q_{\text{glc}}, q_{\text{O}_2})$ signature census, full 34×34 grid (deposited census grid)
25,107	cut events in the random-cut pool, 4,000 cuts (stabilization artifact)
1,516	iML1515 genes, single knockouts (deposited knockout file)
2,779	double-knockout pairs, five panels (deposited knockout file)

B Proofs

This appendix proves the results of the manuscript in full. Where an argument would repeat textbook material, the standard source is cited in lieu of reproduction; the two long technical arguments (Lemma 2.10 and the reconstruction part (iv) of Theorem 3.1) are delegated to Appendix C and cited there.

B.1 Proof of Proposition 2.6

Write the primal as $\max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine in θ . Its dual is

$$\min \{u(\theta)^\top w^+ - \ell(\theta)^\top w^- : S^\top y + w^+ - w^- = c, w^\pm \geq 0\},$$

with y free. The constraint set $\mathcal{D}$ of the dual does not involve θ , so by weak duality every dual-feasible triple furnishes an affine upper bound on Φ , and by strong duality (feasibility and boundedness are hypotheses) the value equals the dual optimum. The dual objective is linear in (w^+, w^-) for each fixed θ , so the optimum over the polyhedron $\mathcal{D}$ is attained at one of its finitely many vertices. Hence Φ is the pointwise minimum of finitely many affine functions of θ and is therefore concave and piecewise affine (the polyhedral value-function structure is classical (Rockafellar, 1970; Borrelli et al., 2003)).

Since Φ is concave piecewise affine, $-\Phi$ is convex piecewise affine, and for every continuous piecewise-affine function the distributional second derivative is the finite signed facet measure computed in the proof of Proposition 2.7 below; each facet density of $D^2(-\Phi)$ is positive semidefinite because $-\Phi$ is convex — the jump of the gradient across a facet of a continuous convex piecewise-affine function is purely normal, $[\nabla(-\Phi)]_F = \alpha_F n_F$ with $\alpha_F \geq 0$ by monotonicity of the subdifferential. This is the piecewise-affine instance of the Alexandrov Hessian measure of a convex function (Alexandrov, 1939); in particular $D^2(-\Phi)$ is a finite symmetric positive-semidefinite matrix-valued Radon measure. As a measure it is a function of Φ alone, and Φ is a function of the LP data (c, S, ℓ, u) alone: no optimal-flux selection rule enters either object. Every measurable selection v^* of optima reproduces the same $\Phi = c^\top v^*$, so $D^2(-\Phi)$ is selection-invariant; the selection acts only on the flux layer $D^2 v^*$, whose mass moves by ten orders of magnitude between two lexicographic tie-breaks while the value layer stays exactly invariant (the controlled follower-variable experiment, §4.1).

For the GPR structure, recall that under multiplicative activity scaling the capacity of a GPR-constrained reaction is built from gene activities by min (AND) and max (OR) nodes. *AND-only rules*: the capacity $\min_k a_k(\theta) u_j$ is the conjunction of the affine inequalities $v_j \leq a_k(\theta) u_j$, one per participating gene, so the feasible set remains a polyhedron with θ entering only through affine right-hand sides, and the concavity argument above applies verbatim (AND-only subnetworks retain concavity). *Exchange-bound families*: the bounds are affine in θ by assumption; the same argument applies. *Single-gene axes*: hold all activities at the reference level and vary one gene. AND gates fall under the previous case. For an OR gate with further genes held at reference activity, the branch $\max(a_g(t), 1) u_j$ is dominated by the constant reference branch for $a_g(t) \leq 1$ — the knockdown direction of the empirical trajectories — so the bound is constant; if the varying gene is the gate’s only member, the bound is affine. Either way the axis restriction is a parametric LP with affine or constant bounds, hence concave. *Failure under OR*: an OR gate with two genes, capacity $\max(\theta_1, \theta_2)$, all else unbounded, gives $\Phi(\theta_1, \theta_2) = \max(\theta_1, \theta_2)$, which is convex and concave only if affine: at $\theta_A = (1, 0)$ and $\theta_B = (0, 1)$ the midpoint $(\frac{1}{2}, \frac{1}{2})$ violates concavity by exactly $\frac{1}{2}$ (the canonical measured violation; the random off-diagonal two-gene probe violates by 0.548). *OR-containing rules*: the value function is the maximum of finitely many concave piecewise-affine functions, one per branch selection, hence continuous piecewise affine, and it carries the signed crease measure of Proposition 2.12; only the Monge–Ampère layer (Propositions 2.7 and 2.8) requires the concave class. $\square$

B.2 Proof of Proposition 2.7

(i). Let $\{\tau\}$ be the finite cell decomposition on which f is affine, with cell gradients $g_\tau = \nabla f|_\tau$. For a test function $\varphi \in C_c^\infty(\Omega)$ and indices i, j , the definition of distributional derivatives and a cellwise Green identity give

$$\langle \partial_i \partial_j f, \varphi \rangle = \int_\Omega f \partial_i \partial_j \varphi \, d\text{vol} = - \sum_\tau (g_\tau)_j \int_{\partial\tau} \varphi (\nu_\tau)_i \, d\mathcal{H}^{p-1},$$

where ν_τ is the outward normal of τ ; the boundary terms on $\partial\Omega$ vanish for φ compactly supported. On an interior facet F shared by cells τ^+, τ^- the two outward normals are opposite, so the facet contributes $-([g]_F)_j (n_F)_i \int_F \varphi \, d\mathcal{H}^{p-1}$ with $[g]_F = g_{\tau^+} - g_{\tau^-}$ and $n_F = \nu_{\tau^+}$; the jump is constant along F because f is affine on each cell. Since f is continuous, the tangential components of the gradient agree across F (both equal the tangential derivative of the common trace), so $[g]_F$ is purely normal, $[g]_F = [\partial_n f]_F n_F$, and the facet density $(g^+ - g^-) \otimes n_F = [\partial_n f]_F n_F \otimes n_F$ is symmetric and constant on F . Hence the singular part of $D^2 f$ is the finite facet sum $\sum_F [\partial_n f]_F n_F \otimes n_F \mathcal{H}^{p-1}|_F$. There are no atoms: each facet F incident on a codimension-two vertex v meets $B_\varepsilon(v)$ in $(p-1)$ -measure at most $C\varepsilon^{p-1}$ (a ball of F 's affine hull), and there are finitely many facets, so the total variation mass in $B_\varepsilon(v)$ is $O(\varepsilon^{p-1}) = O(\varepsilon)$ in the plane — the measured log–log slope 1.000.

(ii). For convex f the Monge–Ampère measure is defined by $\nu_f(E) = |\partial f(E)|$, the Lebesgue measure of the image of the subdifferential (Gutiérrez, 2001). At an interior cell point the subdifferential is the single gradient; at a facet point it is the segment $\text{conv}\{g^+, g^-\}$, of area zero; at a codimension-two vertex v the subdifferential is the convex hull of the gradients of the affine pieces active at v (the max rule for subdifferentials, Rockafellar, 1970), which for three or more active pieces in the plane is a two-dimensional polytope — the normal fan of v . Hence $\nu_f(\{v\}) = |\text{conv}\{n_i\}|$, the volume of the normal fan, and these are the only atoms, since facet-level subdifferentials carry zero two-dimensional area. The measured constancy of the atom under ε -shrinking of the vertex neighborhood is the $\varepsilon \rightarrow 0$ limit of this identity (0.5 for the canonical right-triangle fan). $\square$

B.3 Proof of Proposition 2.8

Let $g = -\Phi$, convex piecewise affine (Proposition 2.6). The minimum of a linear objective over the dual polyhedron $\mathcal{D}$ is attained at a vertex, so g is the maximum of the finitely many affine dual objectives $C_i(\theta) = -(u(\theta)^\top w_i^+ - \ell(\theta)^\top w_i^-)$ indexed by the basic dual solutions $(\lambda_i, y_i, w_i^\pm)$; with the parameters carried into the upper bounds as $u(\theta) = u_0 + U\theta$, the gradient of the i -th piece is $\nabla C_i = -U^\top y_i$ (the bound prices y_i are the sensitivity coefficients of the value function — Danskin's theorem (Danskin, 1967)). The pieces active at θ_v are exactly those of the basic dual solutions optimal at θ_v , and the dual optimal face is the face of $\mathcal{D}$ on which the θ_v -objective attains the optimum, whose vertices are precisely those basic optimal solutions (Ziegler, 1995; Borrelli et al., 2003).

By the max rule (Rockafellar, 1970), $\partial g(\theta_v) = \text{conv}\{\nabla C_i : i \text{ active}\}$, the normal fan of g at θ_v ; and by the subdifferential-image formula for the Monge–Ampère measure (proof of Proposition 2.7(ii); Gutiérrez, 2001),

$$\det D^2(-\Phi)(\{\theta_v\}) = \nu_g(\{\theta_v\}) = |\partial g(\theta_v)| = |\text{conv}\{\nabla C_i\}|,$$

the area of the normal fan: the first identity of the display. For the second, the linear map $y \mapsto U^\top y$ sends the dual optimal face to the polytope $\text{conv}\{U^\top y_i\} = -\text{conv}\{\nabla C_i\}$ (linear maps commute with convex hulls), so the two polygons are reflections of each other through the origin and have equal area:

$$|\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

The machine verification enumerates the dual optimal face directly: vertex refined to machine precision by the exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 against dual-face enumeration area 0.235000068 over 16 direction LPs. $\square$

B.4 Proof of Lemma 2.10

The proof is delegated to Appendix C: well-definedness of the staged selections by compactness and coordinate pinning; polyhedral chamber structure of each stage by multi-parametric LP theory (Borrelli et al., 2003); polyhedrality of the coordinate-fixing step by Fourier–Motzkin elimination (Ziegler, 1995); continuity and piecewise affinity of the inverse of the injective polyhedral projection. $\square$

B.5 Proof of Theorem 2.11

$\Phi = c^\top v^*$ holds pointwise by optimality. Both sides are continuous piecewise affine (Lemma 2.10 for v^* ; LP value function theory for Φ), and differentiation of distributions is linear: applying D^2 to the pointwise identity gives the measure identity $D^2\Phi = \sum_r c_r D^2v_r^*$. Restricting to a crease facet F gives the per-crease form $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$ at every crease time t_k of any piecewise-affine trajectory, and the contraction bound $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$ is Hölder’s inequality. (i) An event with unique optima on both sides is a transversal optimal-vertex switch; its slope jump satisfies $c^\top \Delta v' = \Delta\Phi' \neq 0$ unless the value function is affine through it, which would force the optimum to persist — hence objective-invisibility ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) occurs exactly inside ≥ 1 -dimensional optimal faces, where the tie-break selects among vertices without moving the objective; Theorem 2.11(ii) is the sparse-objective special case. The regularity input (continuity and piecewise affinity of v^*) is measured, not assumed: segment residuals $\leq 8 \times 10^{-14}$ (§4.1, glucose sweep) and Φ piecewise affine at 4.2×10^{-13} (§4.4). $\square$

B.6 Proof of Proposition 2.12 (collapse)

If f is λ -semiconvex, then $D^2f + \lambda I \cdot \text{Leb} \succeq 0$ as symmetric-matrix measures. For continuous piecewise-affine f , D^2f is supported on the codimension-one skeleton, hence purely singular with respect to Lebesgue; testing the inequality on skeleton sets forces the singular part itself PSD, so $D^2f \succeq 0$. Mollifying, $D^2(f * \rho_\varepsilon) = (D^2f) * \rho_\varepsilon \succeq 0$, and $f * \rho_\varepsilon \rightarrow f$ uniformly, so f is convex as a uniform limit of convex functions. The converse is trivial. The numerical form: the smallest violating scale $h_{\max}(\lambda)$ of the OR-plus-cap value function satisfies $h_{\max} = 1/(2\lambda)$ — i.e. $\lambda h_{\max} = 1/2$, the measured 0.500 identity of the OR-plus-cap value function below — exactly, i.e. no finite semiconvexity constant exists at any scale. The signed-layer statement is elementary distribution theory: a continuous PWL function has distributional D^2 supported on its crease skeleton, a finite symmetric Radon measure with per-crease sign flags. $\square$

B.7 Proof of Proposition 2.13

At a generic codimension-two vertex of a convex PWL f on $\mathbb{R}^2$, exactly three facets meet with gradients n_1, n_2, n_3 (cyclically ordered). The Monge–Ampère atom is the volume of the normal fan $\text{conv}\{n_i\}$ (the gradient image sweeps the triangle). Writing $j_i = n_{i+1} - n_i$ for the codimension-one jumps, twice the fan area is $|\det(j_1, j_2)|$ (the shoelace formula on the triangle), giving atom $= \frac{1}{2} |\det(j_1, j_2)| = \frac{1}{2} |j_1| |j_2| \sin \angle$; the product $|j_1| |j_2|$ overestimates by $2/\sin \angle \geq 2$ with equality iff $j_1 \perp j_2$. The simplex-fan generalization to dimension p is the same shoelace argument on a p -simplex fan. Machine verification: 400 random max-of-affine vertices, determinant = fan area to 8.9×10^{-16} , the $2/\sin \angle$ law to 2.5×10^{-12} , orthogonal edge case exactly 2. $\square$

B.8 Proof of Theorem 3.1

(i). The interpolant u_h is continuous piecewise affine, so the facet computation of the proof of Proposition 2.7(i) applies verbatim: $D^2 u_h$ is the finite signed measure $\sum_F [\nabla u_h]_F \otimes n_F \mathcal{H}^{p-1}|_F$ over the interior facets of $\mathcal{T}_h$, with the jump constant on each facet (hence the atom $A_F = [\nabla u_h]_F \otimes n_F |F|$ of Definition 2.1, and $[\nabla u_h]_F = [\partial_n u_h]_F n_F$ by continuity of u_h). No atoms sit on lower-dimensional strata: the mass in $B_\varepsilon(v)$ around a point v of a codimension-two stratum is at most $\sum_{F \ni v} |[\partial_n u_h]_F| \mathcal{H}^{p-1}(F \cap B_\varepsilon(v)) = O(\varepsilon^{p-1}) \rightarrow 0$, each facet being a $(p-1)$ -polytope whose intersection with the ball is contained in a $(p-1)$ -ball of radius ε .

(ii). For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2 u_h, \varphi \rangle = \int_\Omega u_h D^2 \varphi \, \text{dvol}$ by the definition of distributional derivatives; no boundary terms arise because φ is compactly supported. The interpolation bound $\|u - u_h\|_\infty \leq \frac{1}{2} h^2 |u|_{C^2}$ holds on shape-regular families by a barycentric argument: for $x \in \tau$ with vertices z_i and barycentric coordinates $\lambda_i(x)$, $u_h(x) - u(x) = \sum_i \lambda_i(x)(u(z_i) - u(x))$, and Taylor's formula at x gives $u(z_i) - u(x) = \nabla u(x)^\top (z_i - x) + R_i$ with $|R_i| \leq \frac{1}{2} |u|_{C^2} |z_i - x|^2$; the linear terms sum to $\nabla u(x)^\top (\sum_i \lambda_i(x) z_i - x) = 0$ (the barycenter identity), leaving $\|u_h - u\|_\infty \leq \frac{1}{2} h^2 |u|_{C^2}$. Hence $\int (u_h - u) D^2 \varphi \rightarrow 0$ and $\langle D^2 u_h, \varphi \rangle \rightarrow \int u D^2 \varphi = \langle \mu, \varphi \rangle$. For the flat-norm statement, the Kantorovich–Rubinstein norm is the dual norm over the class $\{\phi : \text{Lip}(\phi) \leq 1, \|\phi\|_\infty \leq 1\}$ (Villani, 2009); on the compact Ω this class is uniformly equicontinuous and uniformly bounded, hence totally bounded in the uniform norm (Arzelà–Ascoli). Weak convergence holds against each element of a uniform ε -net of the class by the first part of this item, and the uniform mass bound of part (iii) controls the remainder of the class, so the supremum over the whole class converges: $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$, componentwise and hence entrywise. No convergence of total variation is used or needed, consistent with part (iii).

(iii). For the upper bound, the gradient of the linear interpolant satisfies, on each simplex τ and for every $x \in \tau$,

$$\nabla u_h|_\tau - \nabla u(x) = \sum_i R_i(x) \nabla \lambda_i, \quad |R_i(x)| \leq \frac{1}{2} h^2 |u|_{C^2(\tau)},$$

a decomposition obtained from the same barycentric representation, using $\sum_i \nabla \lambda_i = 0$ and $\sum_i (z_i - x) \nabla \lambda_i^\top = I$; shape regularity gives $|\nabla \lambda_i| \leq C/h$, so $\|\nabla u_h - \nabla u\|_{L^\infty(\tau)} \leq Ch |u|_{C^2}$. At an interior facet F with midpoint m , $|[\nabla u_h]_F| \leq 2Ch |u|_{C^2}$, and $|F| \leq Ch^{p-1}$; quasi-uniformity bounds the number of facets by $C|\Omega| h^{-p}$. Summing the entrywise norms of the atoms, $\|\mu_h\|_{\text{TV}} \leq C |u|_{C^2} |\Omega|$, uniformly in h . For the liminf, the total variation of a matrix-valued measure is the sum of the total variations of its entries, each lower semicontinuous under weak convergence (Billingsley, 1999), so $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$. That convergence of the total variation itself fails in general is the content of Remark 3.2 (exact ratios $3 - 1/n$ and $1.4 - 1/n$; generic smooth maps 3.6–4.4, including convex data). In the one-signed regime along a one-dimensional family the telescoping identity restores it: for convex $u \in C^2$ on $[a, b]$ with mesh $a = t_0 < \dots < t_N = b$, the interpolant u_h is convex, its slope process u'_h is nondecreasing piecewise constant, and

$$\|\mu_h\|_{\text{TV}} = \mu_h([a, b]) = \sum_k \Delta u'_h(t_k) = u'_h(b) - u'_h(a) \rightarrow u'(b) - u'(a) = \mu([a, b]) = \|\mu\|_{\text{TV}},$$

the endpoint slopes being one-sided secant slopes of u ; the concave case is identical up to sign. The parametric-LP value function is concave, so its one-dimensional restrictions sit in this regime, as does κ^μ along a C^1 trajectory.

(iv). The reconstruction statement is proved in Appendix C (quadratic exactness of the per-cell facet sum on structured complexes, a third-order Taylor remainder bound, and the L^1 summation over cells).

(v). At fixed h , μ_h is a fixed finite facet sum; for the mollified family and a test ψ , $\int \psi \, \text{d}(\mu_h * \varphi_\sigma) = \int (\psi * \tilde{\varphi}_\sigma) \, \text{d}\mu_h \rightarrow \int \psi \, \text{d}\mu_h$ as $\sigma \rightarrow 0$, because $\psi * \tilde{\varphi}_\sigma \rightarrow \psi$ uniformly and μ_h is a finite measure: the limit is the atomic measure. For the window, fix σ and let $h \rightarrow 0$ first:

$$\mu_{h,\sigma}(x) = \langle \mu_h, \varphi_\sigma(x - \cdot) \rangle \rightarrow \langle \mu, \varphi_\sigma(x - \cdot) \rangle = (\mu * \varphi_\sigma)(x),$$

uniformly in x over compact sets, by the weak convergence of part (ii) applied to the family of translates $\{\varphi_\sigma(x - \cdot)\}$, which is uniformly bounded and equicontinuous in C^2 , together with the uniform mass bound of part (iii). The second limit is classical mollification of the smooth density: $\|\mu * \varphi_\sigma - D^2u\|_{L^1} \leq C\sigma|u|_{C^3}$. Combining the two: at matched resolution, with h/σ small enough that the first error is dominated and σ/L_{var} small enough that the second is, the coarse-grained family $\mu_{h,\sigma}$ tracks the smooth curvature density, while for $\sigma \ll h$ it resolves the individual atoms and for $\sigma \gtrsim L_{\text{var}}$ it averages out the very variation it is supposed to measure: the smooth object exists only inside the window $h \ll \sigma \ll L_{\text{var}}$, as measured in §4.3. $\square$

B.9 Proof of Proposition 3.3

The proof combines three facts. (a) *Exact polyhedral identity*. For a polyhedral surface, the angle defect at a vertex equals the area of the spherical image of its normal fan: transporting the incident facet normals to the unit sphere, the spherical polygon they span has area 2π minus the sum of the exterior angles at the vertex, which is the defect (Regge, 1961; Cheeger et al., 1984); the identity is machine-verified to 0.0 on the canonical fan (defect, spherical excess, and fan integral all equal). (b) *Smooth graph identity*. For the graph $S_u = \{(x, u(x))\}$, the Gauss map $n(x) = (-\nabla u(x), 1)/\sqrt{1 + |\nabla u|^2}$ has spherical Jacobian per unit *projected* (parameter) area

$$|\text{Jac } n| = K \frac{dA}{\text{dvol}} = \frac{\det D^2u}{(1 + |\nabla u|^2)^{3/2}},$$

because the Jacobian with respect to *surface* area is the Gaussian curvature $K = \det D^2u/(1 + |\nabla u|^2)^2$ (the determinant of the shape operator, the classical Gauss-map identity) and the surface element is $dA = \sqrt{1 + |\nabla u|^2} \text{dvol}$. (c) *Convergence*. The vertex defect measure of the interpolant's graph assigns to each parameter-domain vertex the spherical area of its normal fan; by (a) this is the polyhedral Gauss-map image, and the curvature measures of piecewise-flat approximations converge weakly to the smooth curvature measures (Cheeger et al., 1984). On the graph of u_h over the triangulated parameter domain, the facet normals track the smooth Gauss map at scale h (the gradient interpolation bound $\|\nabla u_h - \nabla u\|_\infty \leq Ch|u|_{C^2}$ of the proof of Theorem 3.1(iii)), so the union of the vertex normal fans over a parameter region R covers the spherical image $n(R)$ up to a vanishing boundary discrepancy. Combining (a)–(c), against a test ψ on the parameter domain,

$$\int \psi \, dK_h \rightarrow \int \psi(x) \frac{\det D^2u(x)}{(1 + |\nabla u(x)|^2)^{3/2}} \text{dvol},$$

the Gauss-map area density per projected parameter area. The *intrinsic* Gaussian curvature is the same integrand divided by $\sqrt{1 + |\nabla u|^2}$; the discrepancy factor is exactly $\sqrt{1 + |\nabla u|^2} = \sec(\text{tilt})$, the tilt being the angle between the graph normal and the vertical: the sec law. Recovering intrinsic curvature therefore requires multiplying by the graph area correction, as stated. The 0.0% excess across the 13 probes spanning tilts 0.11–2.83 and three tilt directions is the numerical form of the exact factor. $\square$

B.10 Proof of Proposition 3.5

Smooth case. For a C^2 map the curvature mass is the total variation of D^2u , whose density is the bounded matrix field D^2u : the mass captured by a planar loop of radius ε is at most $\|D^2u\|_{L^\infty}$ times the loop's area $\pi\varepsilon^2$ — $B(\varepsilon) = O(\varepsilon^2)$, the measured slope 2.000 on the smooth arm of the regime-dial design.

Piecewise-affine case. For a piecewise-affine map the mass measure D^2v is the facet-layer measure of the proof of Proposition 2.7(i): finitely many facets, each carrying a bounded density against $(p - 1)$ -dimensional facet measure. A loop of radius ε meets any given facet line in a

chord of length at most 2ε , so the captured mass is at most density times 2ε per facet crossed: $B(\varepsilon) = O(\varepsilon)$. When the kink complex meets the loop *transversally*, the crossing is genuine, the chord is of exact order ε , and the captured mass is $\Theta(\varepsilon)$: the log-log slope 1.00 of the parametric-FBA arm (76-pair perturbation scan). In general parameter dimension p the same facet-measure computation gives $B(\varepsilon) = O(\varepsilon^{p-1})$; the dichotomy as stated is the planar case, which is where it is measured. $\square$

C Technical proofs

The two long arguments deferred from Appendix B.

C.1 Proof of Lemma 2.10

Well-definedness. For each θ the feasible set $P(\theta) = \{v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ is a nonempty compact polytope (hypothesis). Stage 1 maximizes a linear objective over $P(\theta)$; its argmax $P_1(\theta)$ is a nonempty compact face. Stage 2 maximizes the secondary objective over $P_1(\theta)$, and so on. Two selection rules complete the lexicographic stack, and we state both. The *coordinate-pinned completion* minimizes the coordinates $v_1, \dots, v_n$ one at a time, in a fixed order, over the current face: after the coordinate v_j has been fixed to its conditional minimum, the remaining set is the face intersected with the hyperplane $v_j = m_j$; after all n coordinates every component is pinned to a specific value, so the final set is a single point $v^*(\theta)$ — uniqueness is unconditional. The *engine's weighted stage 3* (§7) minimizes $w^\top v$ over the compact stage-2 face with the fixed weight vector w (seed 20240901): for stage-3 objectives of this form the argmin is a unique vertex of the face for every θ outside a Lebesgue-null set of weight vectors (the finitely many hyperplane arrangements $\{w^\top v = w^\top v'\}$, $v \neq v'$ vertices of the face, bound the exceptional set), so the fixed seed realizes a generic draw and the pinned vertex is unique for it; the chamber and projection argument below then applies verbatim, and the coordinate-pinned rule is the representative for which uniqueness holds for *every* w . Existence at every stage follows from compactness, and the lexicographic optimum is therefore well defined under both completions.

Piecewise affinity and continuity. Consider the graph

$$G = \{(v, \theta) : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\} \subset \mathbb{R}^n \times \mathbb{R}^p,$$

a polyhedron, since ℓ and u are affine. The optimal-set graphs are tracked through the stages by induction on the stage index k , with the following claim.

Claim. For each objective stage k there is a finite partition of the parameter box into closed polyhedral chambers $\{K_{k,\alpha}\}$ such that the stage- k value Φ_k is affine on each chamber and the stage- k graph $G_k = \{(v, \theta) \in G_{k-1} : c_k^\top v = \Phi_k(\theta)\}$ intersects $\mathbb{R}^n \times K_{k,\alpha}$ in a polyhedron.

For $k = 1$ this is the structure theorem of multi-parametric linear programming: for an LP whose parameters enter affine right-hand sides, the parameter space admits a finite closed polyhedral chamber cover on which the value is affine and the optimal face varies polyhedrally (Borrelli et al., 2003). Explicitly, over a chamber K with $\Phi_1 = \ell_K$ affine,

$$G_1 \cap (\mathbb{R}^n \times K) = \{(v, \theta) : (v, \theta) \in G, \theta \in K, c_1^\top v = \ell_K(\theta)\},$$

an intersection of polyhedra; the affine constraint $c_1^\top v - \ell_K(\theta) = 0$ is legitimate because $\Phi_1 = \ell_K$ on K , so for $\theta \in K$ membership in the stage-1 face is exactly this equation. For the induction step, write each polyhedron $Q = G_k \cap (\mathbb{R}^n \times K_k)$ in the form $\{(v, \theta) : Av \leq b + B\theta\}$ (every affine constraint in (v, θ) can be so rearranged, with the chamber constraint on θ absorbed into the right-hand side). The stage- $(k+1)$ value $\Phi_{k+1}(\theta) = \max\{c_{k+1}^\top v : Av \leq b + B\theta\}$ is again a multi-parametric LP value function, so it subdivides K_k into finitely many closed sub-chambers on which it is affine and the stage- $(k+1)$ graph is polyhedral, exactly as at $k = 1$ (Borrelli et al., 2003). This proves the claim for the objective stages: each value Φ_k is continuous piecewise affine,

and each graph G_k is a finite union of polyhedra, polyhedral over each chamber of a common refinement.

The coordinate-fixing stage keeps polyhedrality by Fourier–Motzkin elimination (Ziegler, 1995). Let $Q = G_k \cap (\mathbb{R}^n \times K)$ be one chamber piece of the stage- k graph and let j be the next coordinate to fix. Consider

$$A_j = \{((v, \theta), (v', \theta')) \in Q \times Q : \theta = \theta', v_j \leq v'_j\},$$

a polyhedron in $\mathbb{R}^{n+p} \times \mathbb{R}^{n+p}$. Its projection onto the first factor is

$$\{(v, \theta) \in Q : v_j \leq v'_j \text{ for every } (v', \theta') \in Q \text{ with } \theta' = \theta\},$$

which is precisely the stage graph after fixing $v_j = m_j(\theta) = \min\{v'_j : (v', \theta') \in Q\}$; projections of polyhedra are polyhedra (Fourier–Motzkin elimination), so the fixed-coordinate graph is polyhedral over the chamber, with compact slices preserved. Iterating over the n coordinates, the final graph G^* is, over each final chamber K_α , a polyhedron that projects injectively onto K_α , the v -slice being the single point $v^*(\theta)$.

Over each final chamber, v^* is the inverse of an injective polyhedral projection. Its continuity is the closed-graph lemma: the graph piece is closed, the slices are singletons, and a function with closed graph and locally bounded values is continuous. Its piecewise affinity is polyhedral geometry: the face lattice of the graph polyhedron projects to a finite polyhedral decomposition of K_α , and on the relative interior of each projected face the inverse map is affine, an injective affine map between polytopes having an affine inverse on its image (Ziegler, 1995). Chambers overlap on shared faces, and the selections computed on different chambers agree there because $v^*(\theta)$ is unique; gluing the affine pieces along the common closed faces gives a globally defined continuous piecewise-affine v^* . The graph over each GPR cell is a finite union of the chamber pieces (the chambers subdivide the cells, since the LP data is affine over a cell), that is, a finite polyhedral complex, as stated. The measured regularity inputs — path segment residuals $\leq 8 \times 10^{-14}$ (§4.1) and Φ piecewise affine at 4.2×10^{-13} (§4.4) — are the numerical form of the exact polyhedral identities above. $\square$

C.2 Proof of Theorem 3.1(iv)

Structured complexes. Fix a translation-invariant complex with period h : each cell C carries a fixed simplex pattern, and the assignment $F \rightarrow C$ of skeleton facets to cells is translation-equivariant. Write $S_C(u) = \sum_{F \rightarrow C} A_F$ for the per-cell signed facet sum and $\rho_h(C) = |C|^{-1} S_C(u)$ for the reconstructed density. Two properties drive the argument.

(a) *Quadratic exactness.* The map $u \mapsto S_C(u)$ is linear, since interpolation is linear in the data; and S_C kills affine functions, since the interpolant of an affine function is exact and no jumps arise. For the quadratic $q_H(x) = \frac{1}{2}x^\top Hx$, scaling homogeneity of the construction (atoms have the form $[\nabla q]|F| \sim h \cdot h^{p-1}$) gives $S_C(q_H) = c(P)h^p H$ with a dimensionless constant $c(P)$ depending only on the cell pattern. For the structured right-triangle lattice, $c(P)h^p = |C|$: the per-cell atoms $A_{\text{diag}}, A_{\text{vert}}, A_{\text{horiz}}$ of Remark 3.2 sum *exactly* to $h^2 H$ for the interpolant of $u = \frac{1}{2}x^\top Hx$ (machine-verified to 0.0). Hence $S_C(q) = |C|D^2 q$ exactly for every quadratic q : for quadratic data the reconstruction is exact, $\rho_h(C) = D^2 q$.

(b) *Third-order remainder.* Let $u \in C^3$ and q its second-order Taylor polynomial at the cell center x_C . On C , $|u - q| \leq Ch^3|u|_{C^3}$ and $|u - q|_{C^2} \leq Ch|u|_{C^3}$. On each simplex of the pattern, the barycentric gradient bound of the proof of Theorem 3.1(iii) gives $\|\nabla(u - q)_h\|_{L^\infty} \leq \|\nabla(u - q)\|_{L^\infty} + Ch|u - q|_{C^2} \leq Ch^2|u|_{C^3}$, so each facet atom of the remainder is bounded by $Ch^2 \cdot h^{p-1} = Ch^{p+1}$, and each cell carries a fixed number of facets: $|S_C(u - q)| \leq Ch^{p+1}|u|_{C^3}$.

(c) *L^1 summation.* By (a) and (b),

$$\rho_h(C) = |C|^{-1}(S_C(q) + S_C(u - q)) = D^2 u(x_C) + O(h|u|_{C^3}),$$

983 using $D^2q = D^2u(x_C)$ and $|C| \sim h^p$. Summing over cells,

$$\begin{aligned} \|\rho_h - D^2u\|_{L^1} &= \sum_C \int_C |\rho_h(C) - D^2u(x)| \, \text{dvol} \\ &\leq \sum_C |C| \left(Ch|u|_{C^3} + \sup_{x \in C} |D^2u(x) - D^2u(x_C)| \right) \leq 2Ch|\Omega| |u|_{C^3}, \end{aligned}$$

984 the second term using $\sup_{x \in C} |D^2u(x) - D^2u(x_C)| \leq h|u|_{C^3}$. This is strong L^1 convergence
 985 at rate $O(h)$ for $u \in C^3$, and consequently $|\|\rho_h\|_{L^1} - \|D^2u\|_{L^1}| \leq \|\rho_h - D^2u\|_{L^1} \rightarrow 0$ with
 986 $\|D^2u\|_{L^1} = \|\mu\|_{\text{TV}}$: the no-mass-loss statement in its correct, cell-averaged form.

987 *Unstructured complexes.* The unweighted three-facet patch of an arbitrary cell is not the
 988 discrete-duality cell: the correct pairing weights each skeleton facet by its covolume (dual cell)
 989 in the normal direction, which is the framework in which the translation-covariance argument
 990 above is replaced by the discrete-duality (dual-diamond) identities (Cheeger et al., 1984). The
 991 theorem therefore asserts the structured case, proved above, and records the pairing required for
 992 the general statement. $\square$

993 Data, Software, and Code Availability

994 All figures, tables, and numerical claims in this manuscript are produced by committed scripts
 995 from deposited artifacts; the complete bundle (generation scripts, result artifacts, genome-scale
 996 models, data manifests with SHA-256 checksums, and the verification scripts that re-derive ev-
 997 ery manuscript number) is available at [github.com/ MIKEAA2020/metabolic-curvature-measure](https://github.com/MIKEAA2020/metabolic-curvature-measure).
 998 Genome-scale models (iJO1366, iML1515) were obtained from the BiGG database; expression
 999 compendia from M3D and PRECISE; proteome data from PaxDb and the condition-dependent
 1000 proteome of Schmidt et al.; all sources are public and cited in the reference list. No new experi-
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 1011 the final content.

1012 Author Contributions

1013 A.A. conceived the study design, developed the methodology and analysis code, performed the
 1014 data analysis, drafted and revised the manuscript.

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